

Lecture 02

Market Failures

Ivan Rudik

AEM 4510

Roadmap

- What are market failures?
- When do they happen?
- What are the consequences?

Market failures and the environment

The ideal world: graphical

In the best case scenario, a market equilibrium leads to the efficient allocation

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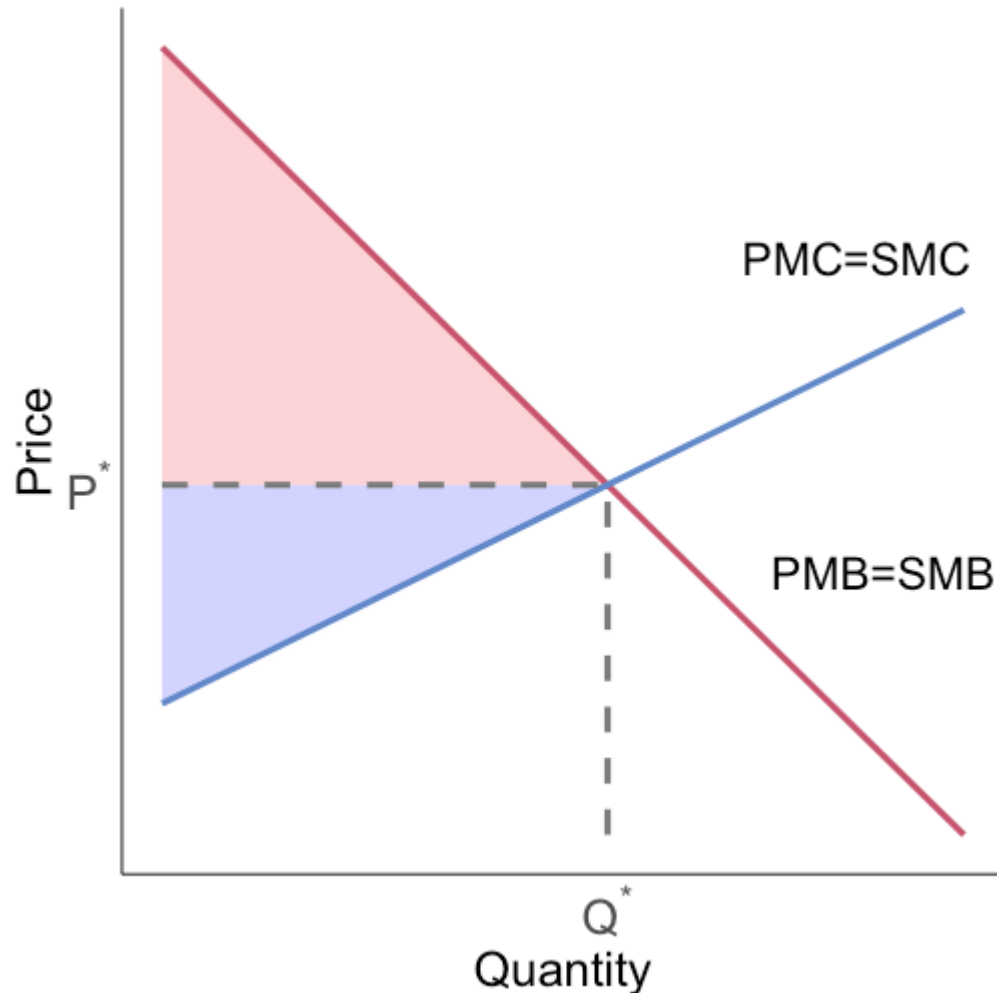
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Market equilibrium

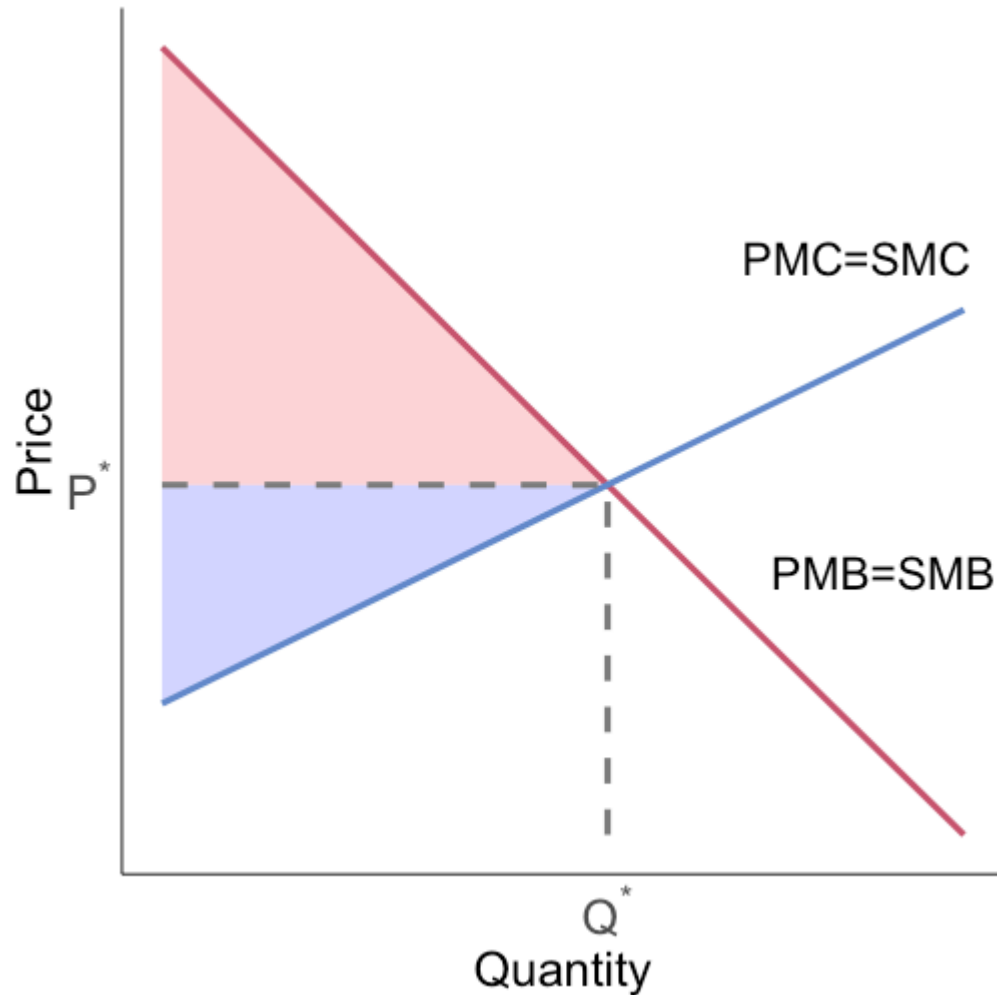


Consumer surplus is the difference between willingness to pay (demand) and price

Producer surplus is the difference between price and marginal cost (supply)

Total surplus is the sum of CS and PS

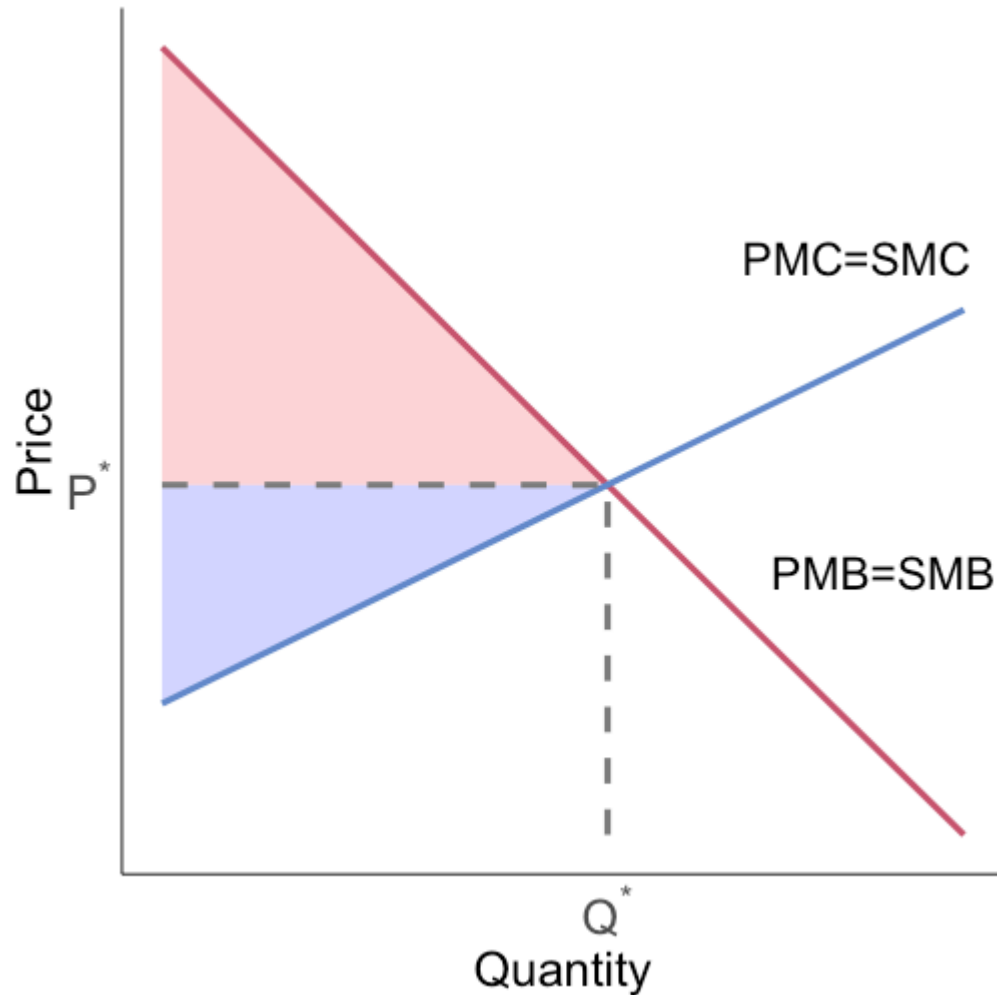
Market equilibrium



For bread, the private costs and benefits are very likely the social costs and benefits

What does this mean about the market allocation?

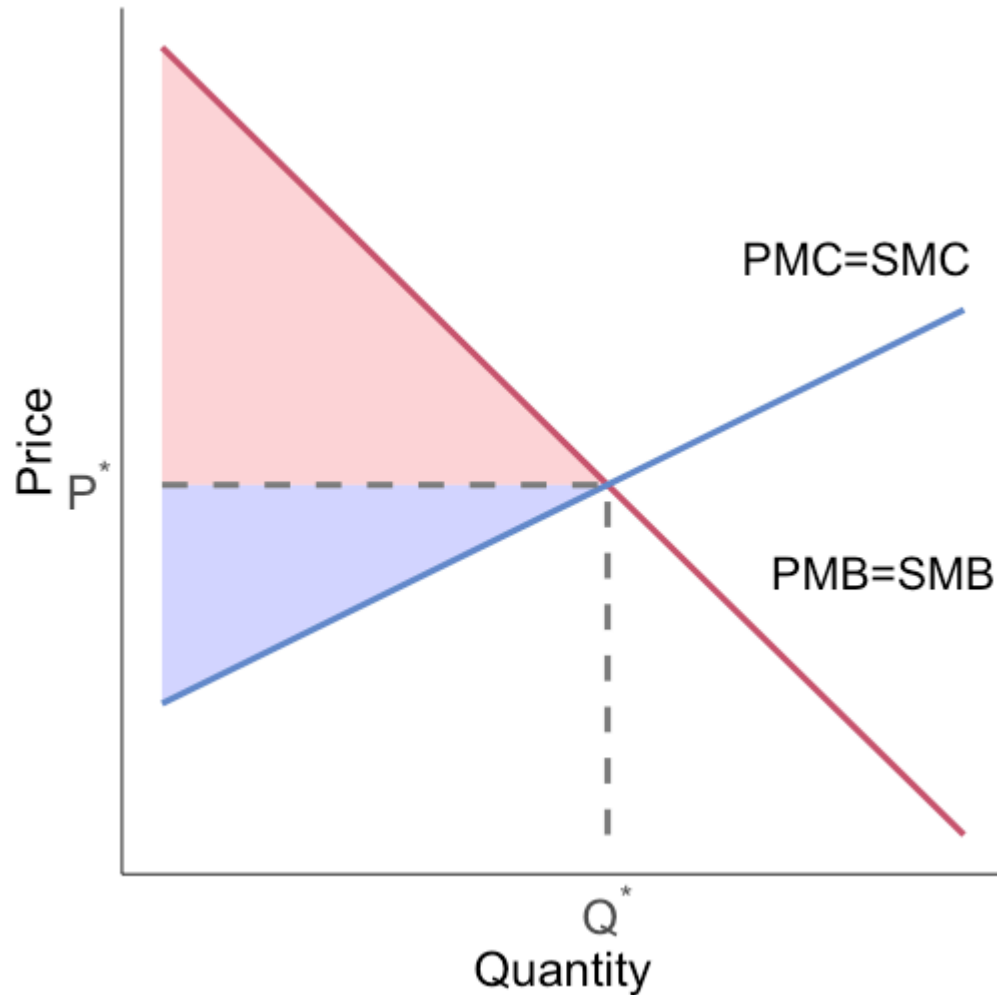
Market equilibrium



The market allocation is **efficient**
because $SMC = SMB$

Why?

Market equilibrium

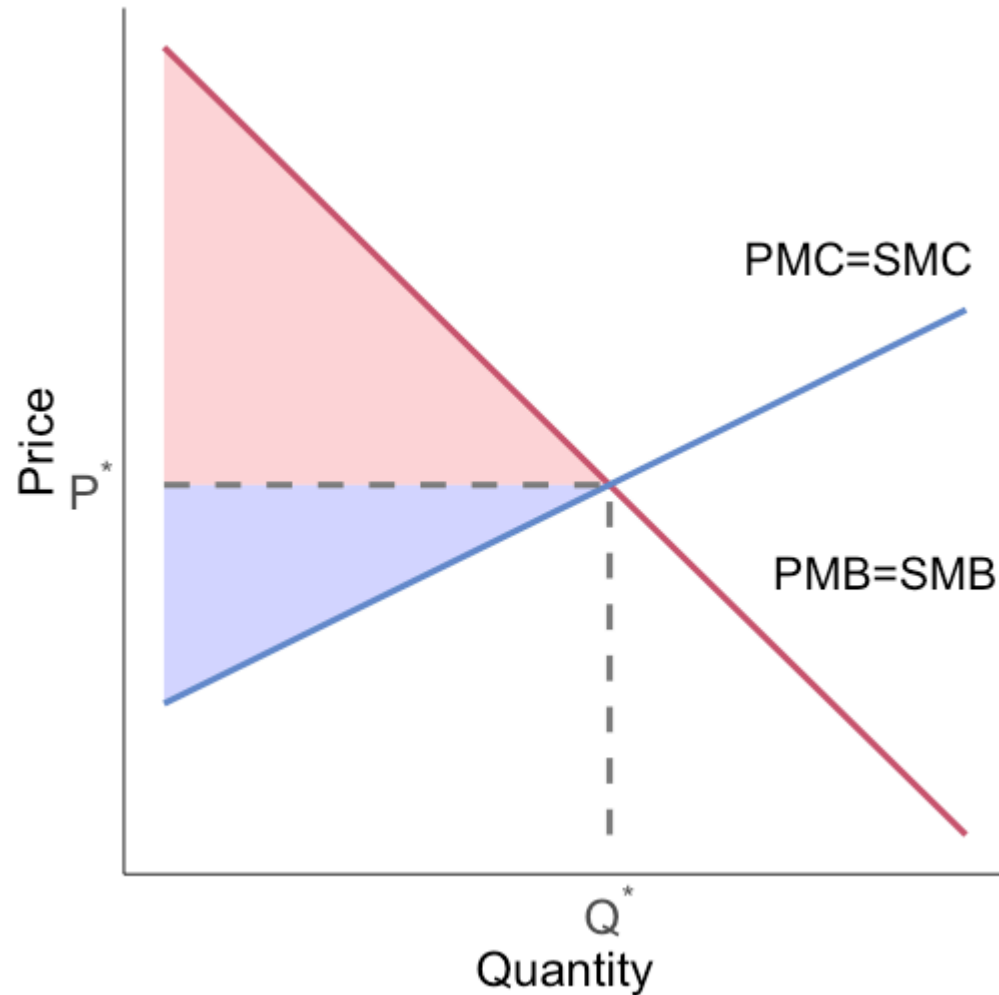


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Why?

Consider deviating from (P^*, Q^*)

Market equilibrium



Cost of next unit after $Q^* >$ benefit

Benefit of last unit $\geq$ cost of last unit before Q^*

Competitive market allocations are efficient for private goods

Externalities

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In the real world we have **externalities**

An externality exists whenever an individual or firm undertakes an action that impacts another individual or firm in an unintended way for which the latter is not compensated (a negative externality), or for which the latter does not pay (a positive externality)

Externalities

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Markets rely on prices to signal the social value of goods

Arizona's water problems share a market-failure structure

Colorado River

- Arizona shares a limited river supply with other states, Tribal Nations, and Mexico
- Each user's withdrawal increases scarcity for other users

Groundwater

- Pumping from a shared aquifer lowers the water table and can cause land subsidence
- Pumpers do not bear all the costs imposed on nearby wells and future users

Both: incomplete property rights + missing prices + high transaction costs

Colorado River rules are being rewritten for 2027



The post-2026 Colorado River regime is not yet settled

As of August 26, 2026:

- Current operating guidelines and drought plans expire at the end of 2026
- The July 31 final environmental impact statement identifies an adaptive framework through 2036
- The Department of the Interior expects final guidelines before October 1

The economic question: who bears shortages when river flows are uncertain?

Saudi farming exposed Arizona groundwater rules



Arizona has begun regulating Ranegras groundwater

- Annual withdrawals have exceeded natural recharge by roughly **900%**
- One monitored well has fallen more than **240 feet** since the 1980s, and groundwater loss has caused land subsidence
- Arizona designated Ranegras Plain as its eighth Active Management Area in January 2026
- The new rules restrict irrigation on new acreage and require large users to measure and report pumping

The change: an open-access aquifer is becoming a managed common-pool resource

Groundwater scarcity is also moving through the courts

- Arizona sued Fondomonte in December 2024, alleging that excessive pumping threatened nearby communities and infrastructure
- In February 2026, the attorney general argued that the new Active Management Area complements rather than replaces the lawsuit
- The state is seeking remedies targeted at the alleged harm, while the Active Management Area regulates pumping throughout the basin

Same externality, different tools: basin-wide regulation and liability for alleged harm

Groundwater enforcement now uses settlements as well as lawsuits



Externalities: several classifications

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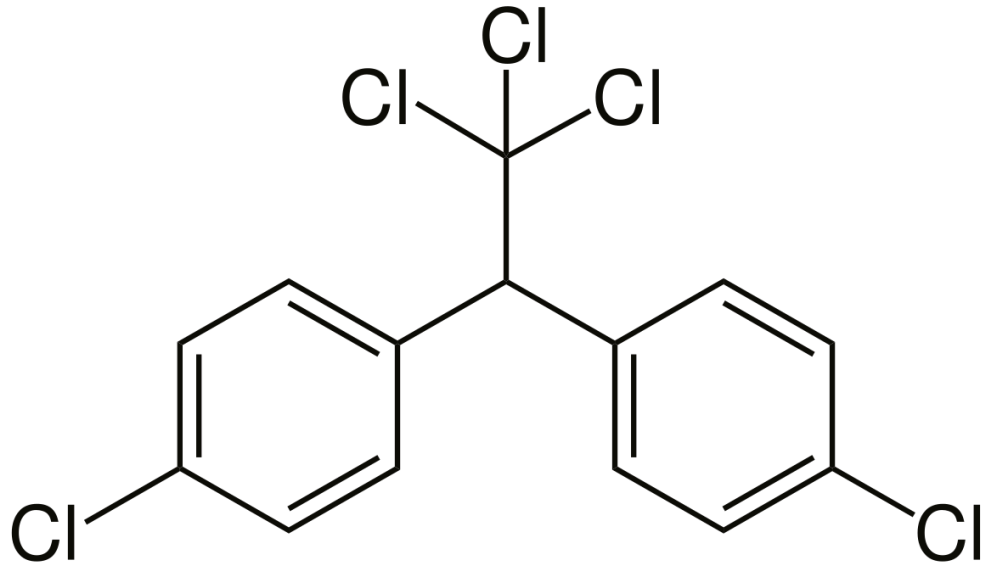
Negative externalities: imposes external **costs** (e.g. pollution)

Positive externalities: imposes external **benefits** (e.g. vaccination)

Negative externalities: what is this?



Negative externalities: DDT, shockingly bad for you

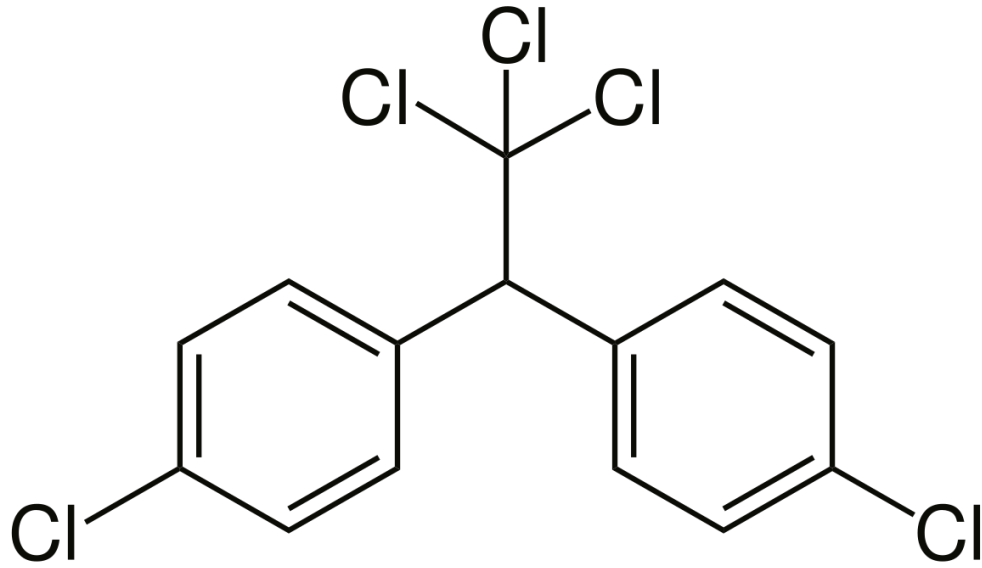


DDT is a chemical that was widely used as an insecticide in the early-to-mid 1900s

Widely used to eradicate Typhus and Malaria

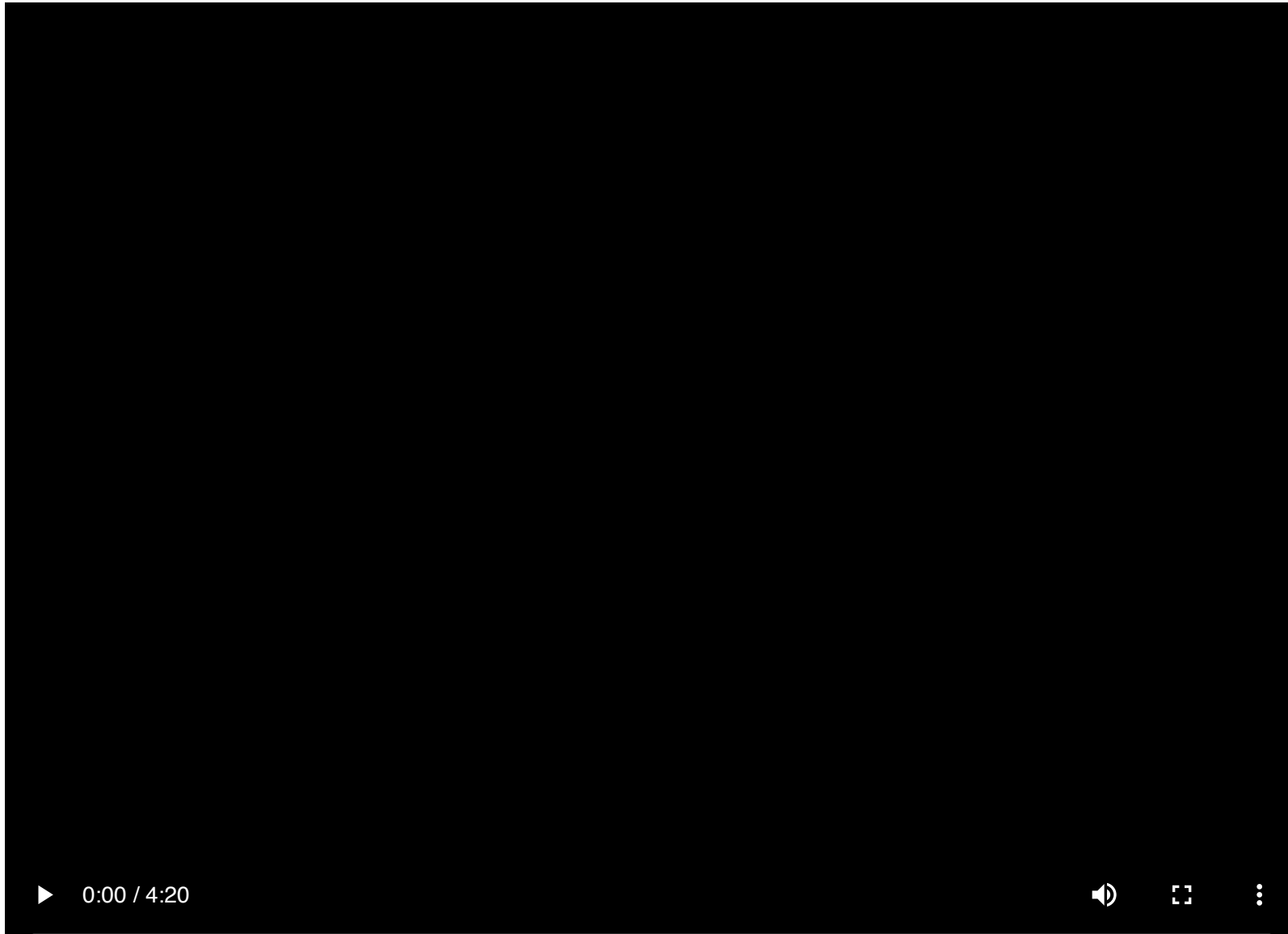
Used to treat lice

Negative externalities: DDT, gives you cancer

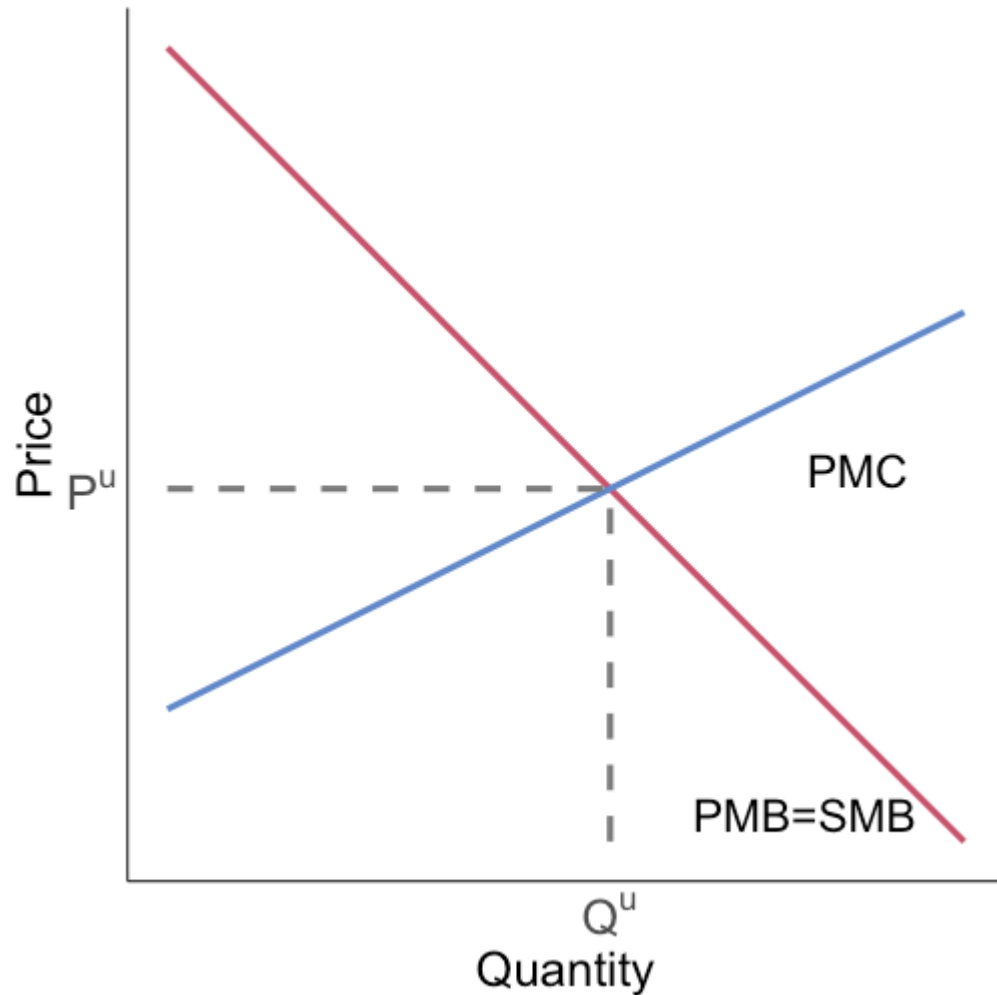


A relationship between DDT exposure and reproductive effects in humans is suspected, based on studies in animals. In addition, some animals exposed to DDT in studies developed liver tumors. As a result, today, DDT is classified as a probable human

The birth of the environmental movement



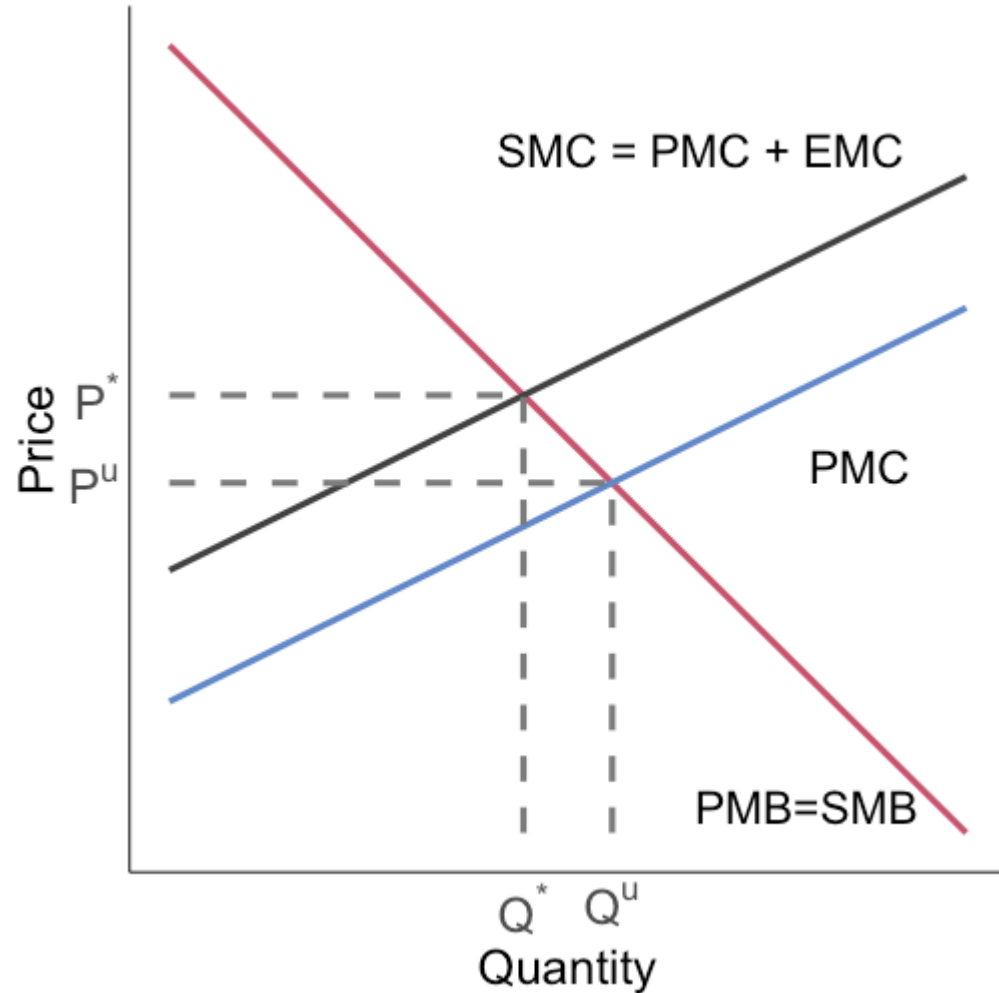
Negative externalities: graphical



Social marginal cost (SMC) is the sum of private marginal cost (PMC) and the external marginal cost (EMC)

Where is the SMC?

Negative externalities: graphical

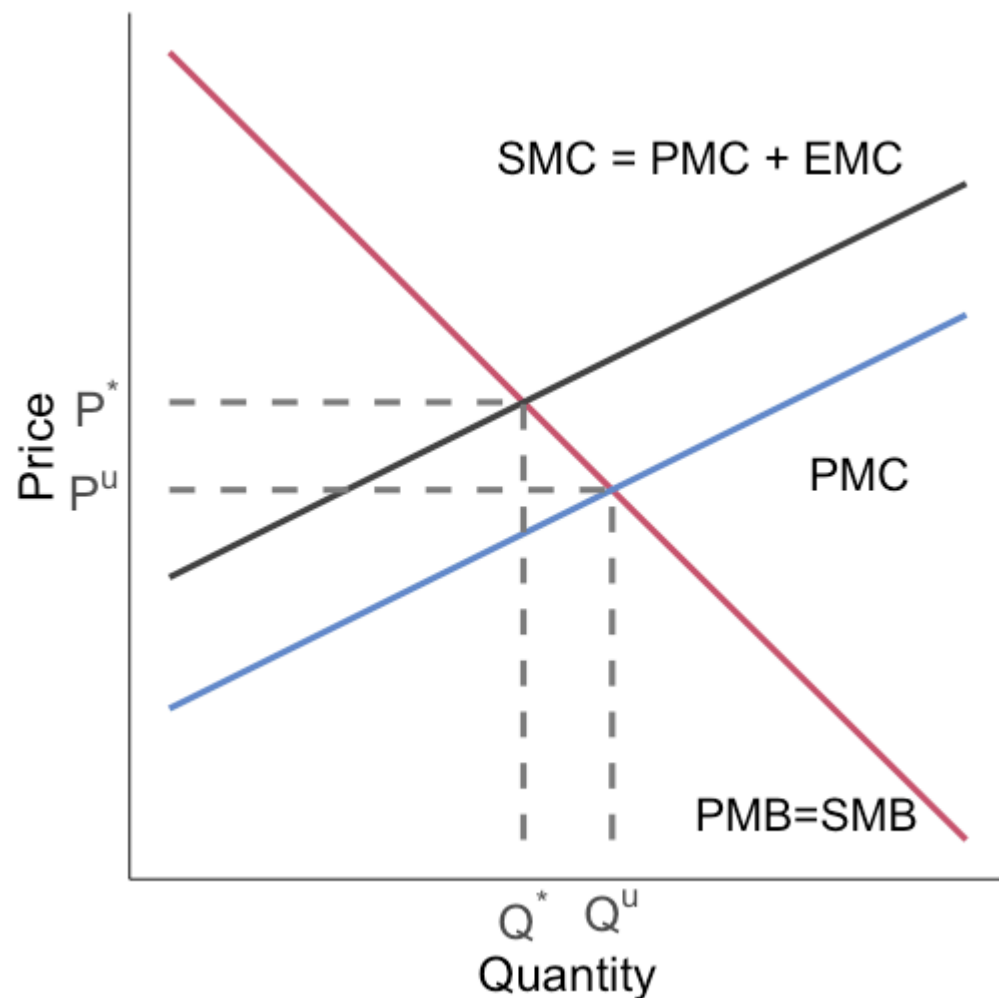


Social marginal cost (SMC) is the sum of private marginal cost (PMC) and the external marginal cost (EMC)

The PMC curve only reflects the **private costs** of making the DDT

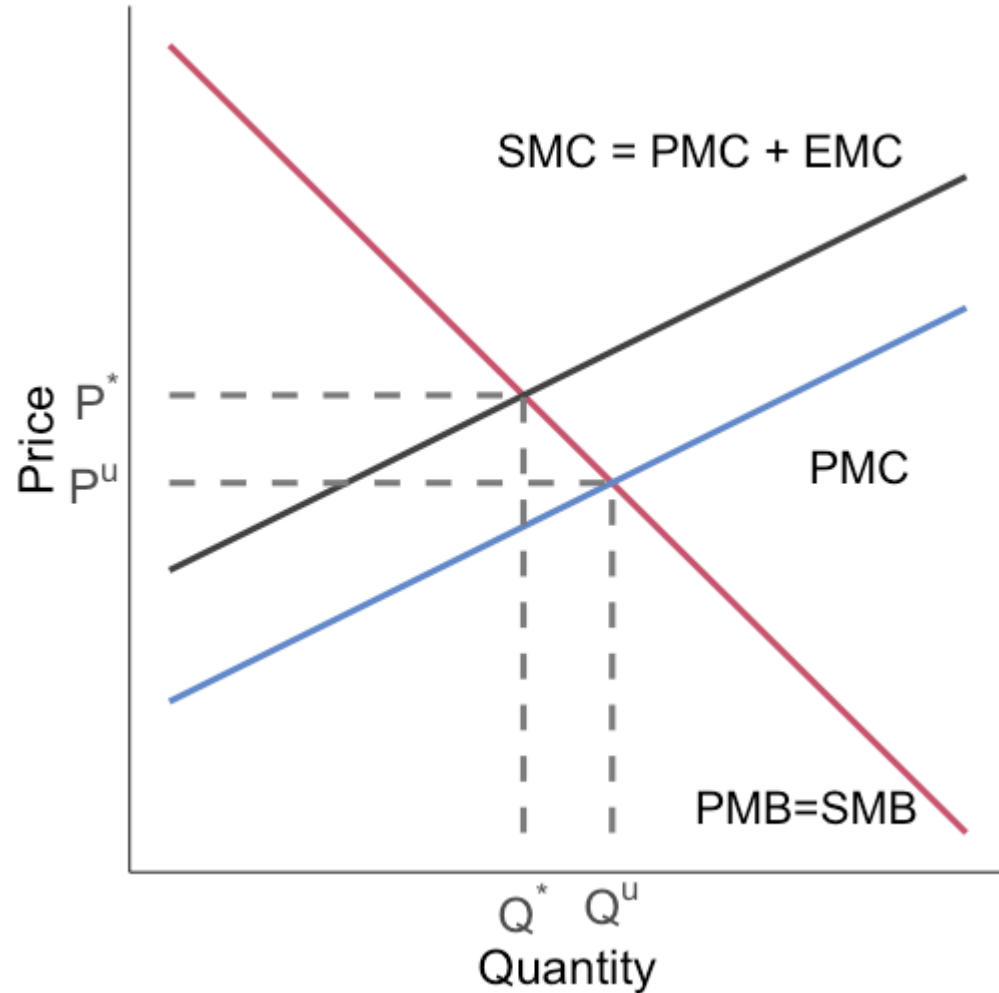
It does not account for the external health and wildlife costs

Negative externalities: graphical



Adding the private and external marginal costs together gives us the SMC, what we care about from the social planner or regulator's perspective

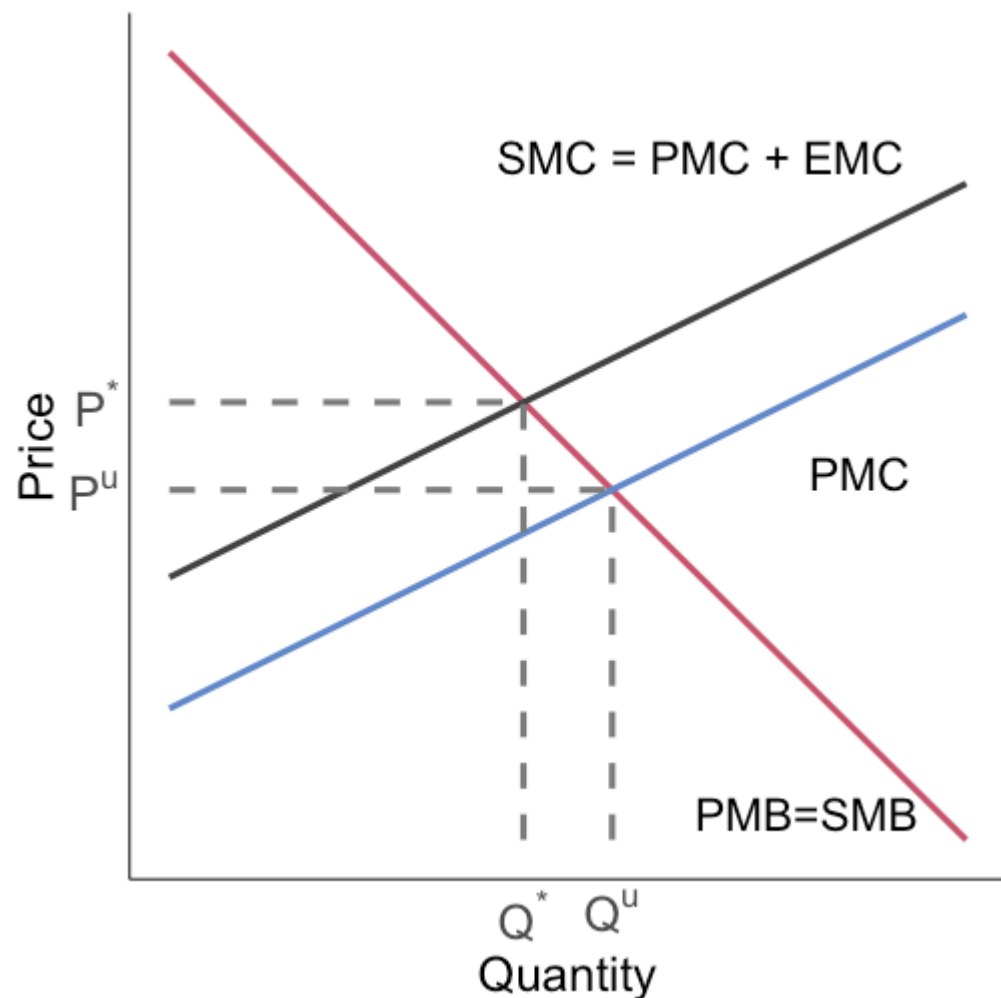
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Negative externalities: graphical



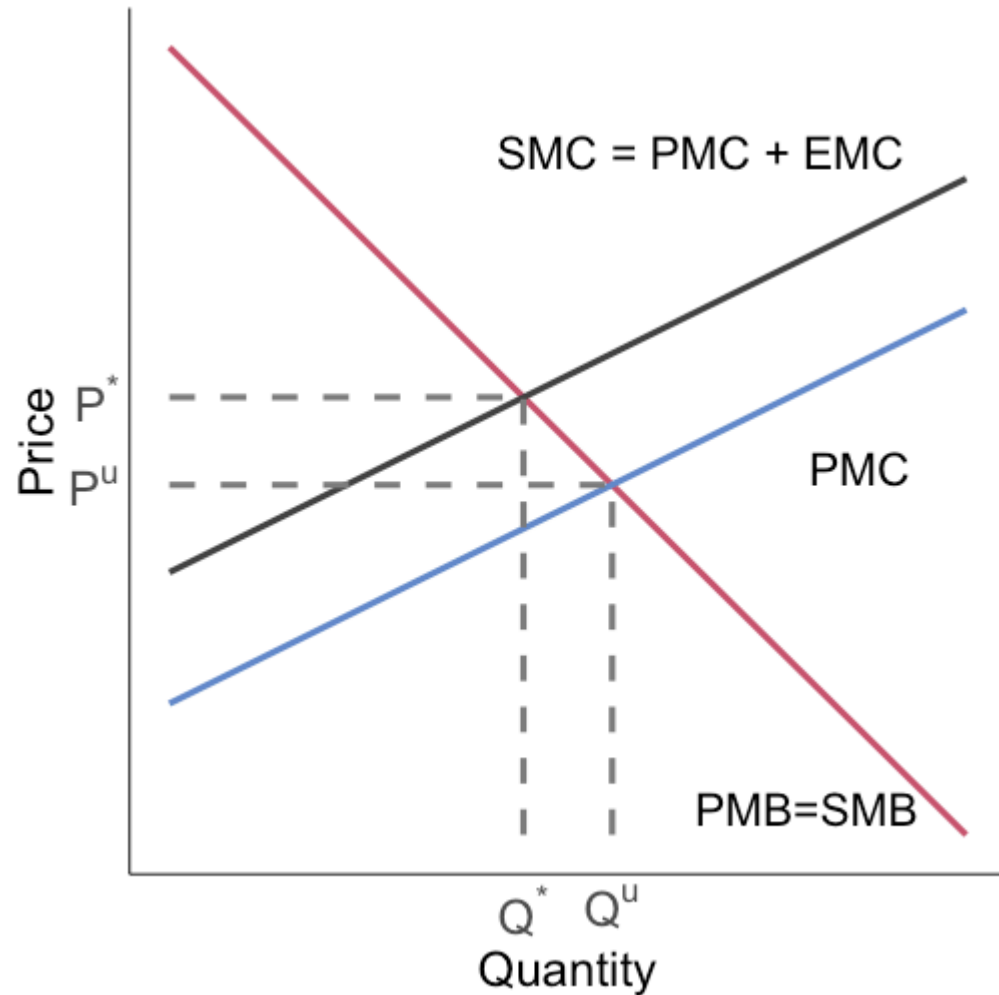
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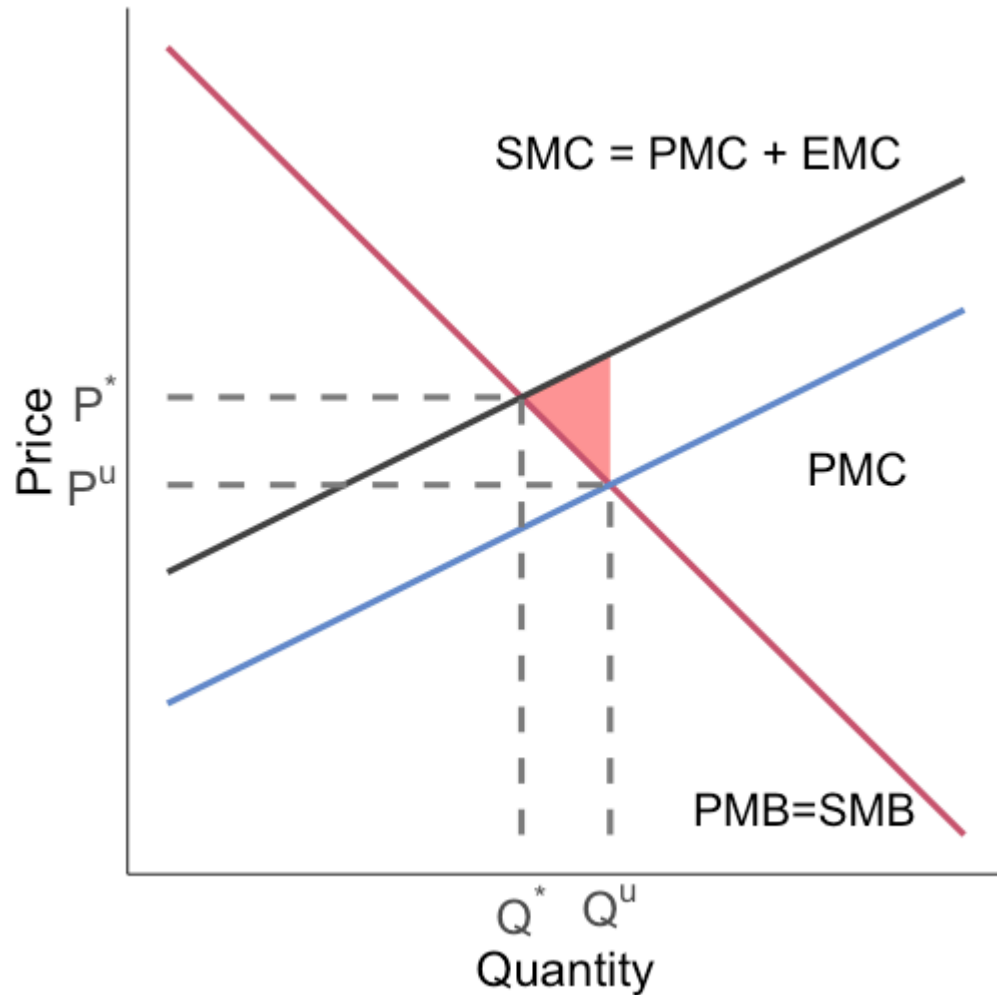
What's the social cost of this market failure?

Negative externalities: graphical

Negative externalities generate deadweight loss equal to...

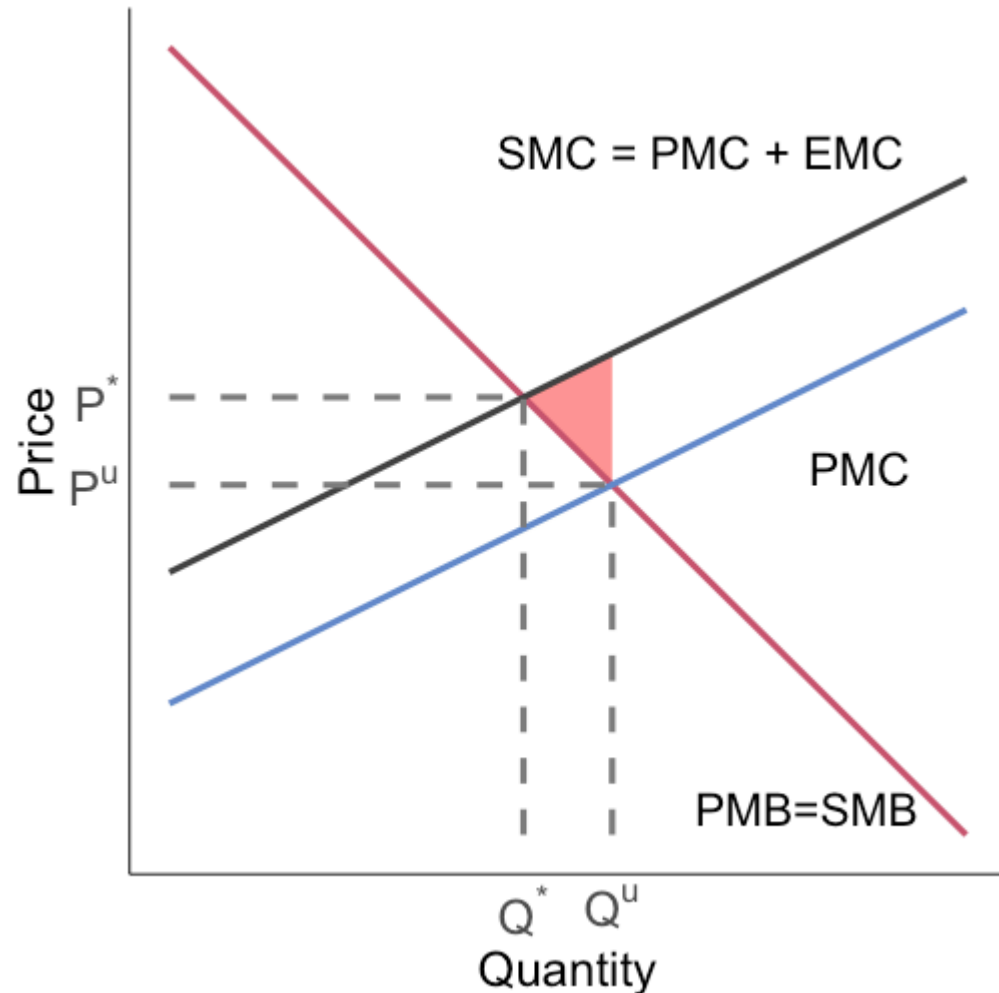


Negative externalities: graphical



Negative externalities generate deadweight loss equal to the **red** area

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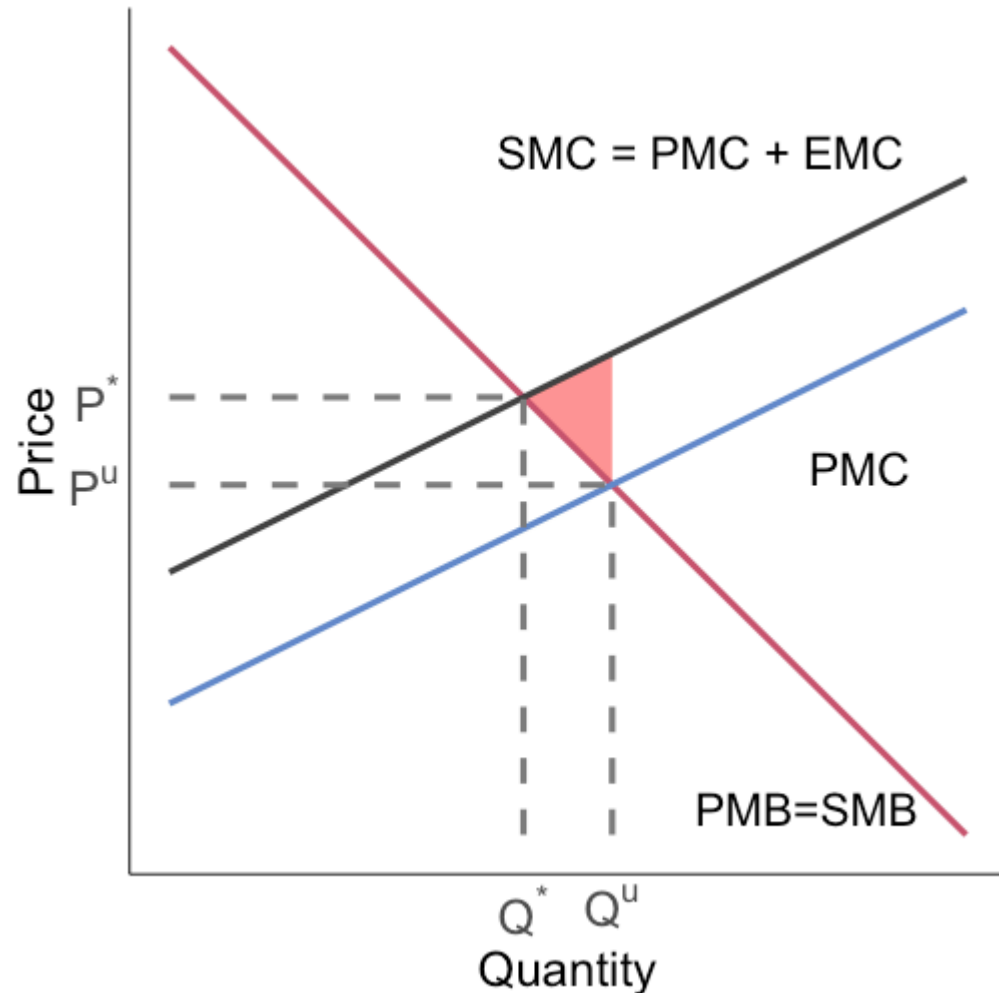


Negative externalities generate deadweight loss equal to the **red** area

This is the difference in SMC and SMB for units bought/sold where $SMC > SMB$:

Total $SMC - SMB$ from Q^* to Q^u

Negative externalities: graphical



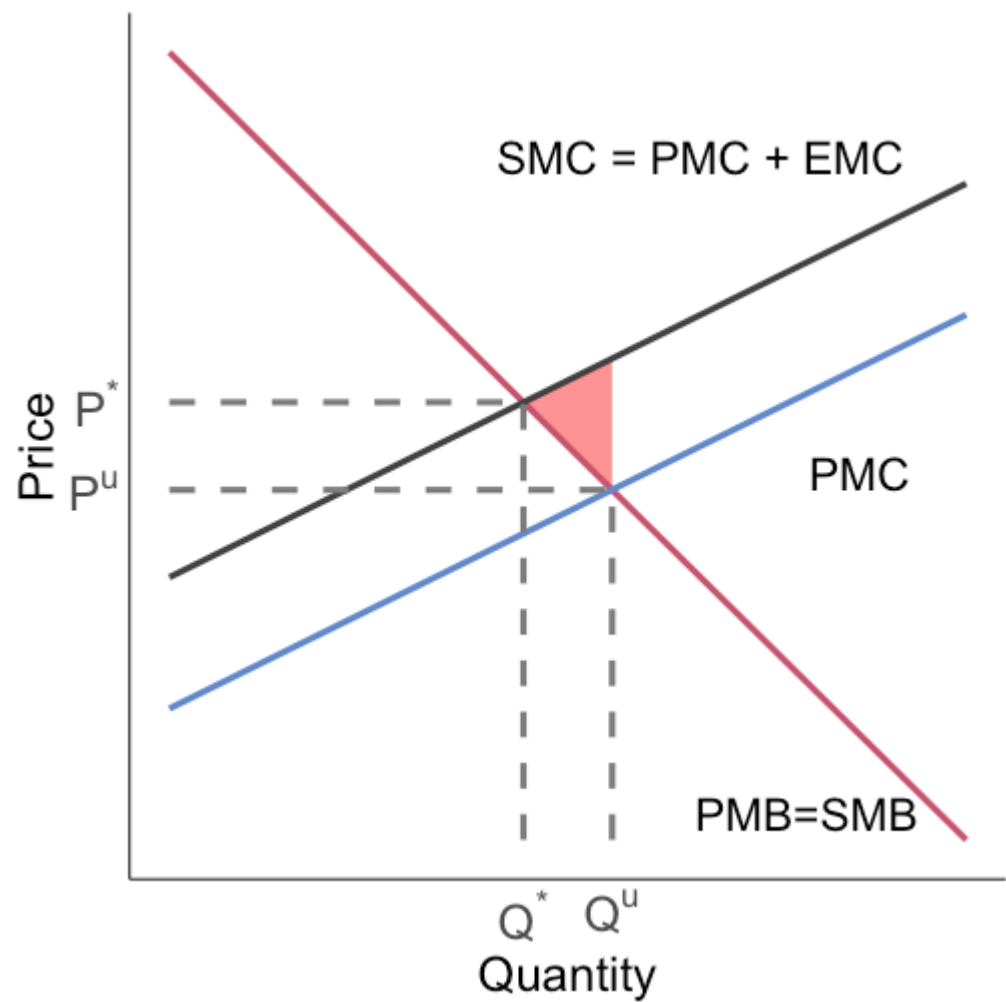
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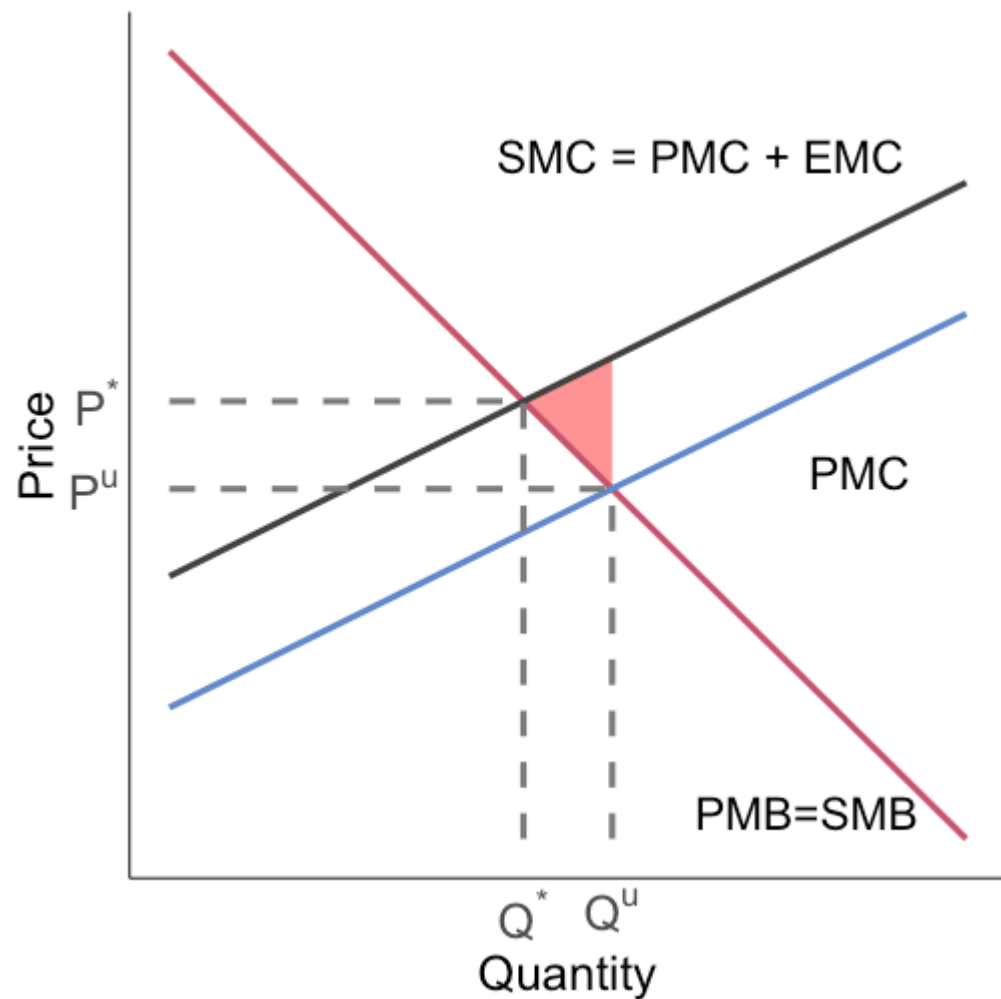
This is the loss to society caused by the externality in the unregulated private market

Negative externalities: graphical

Key takeaway:



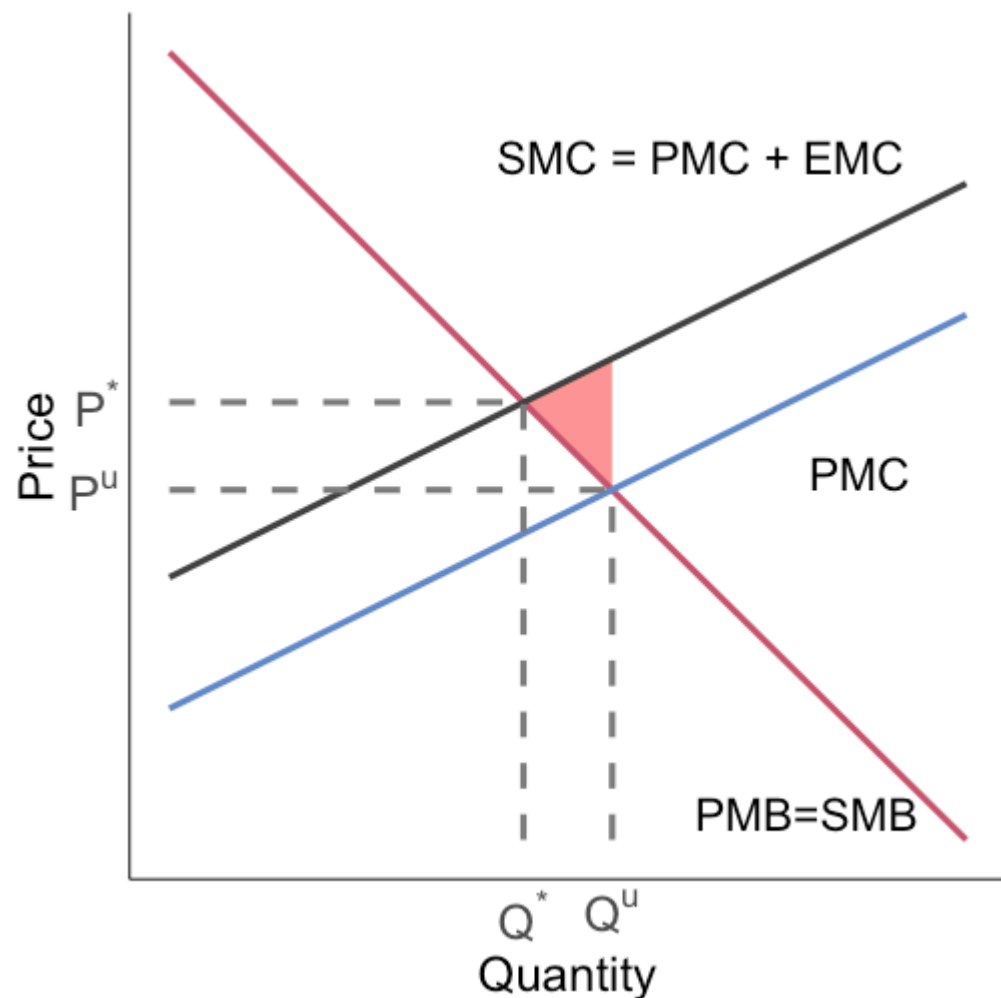
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Key takeaway:

The private market produces too much DDT

Negative externalities: graphical



Key takeaway:

The private market produces too much DDT

The private actors are not accounting for the **external costs** they are imposing on people who are not in the DDT transaction (e.g. third parties whose health is affected)

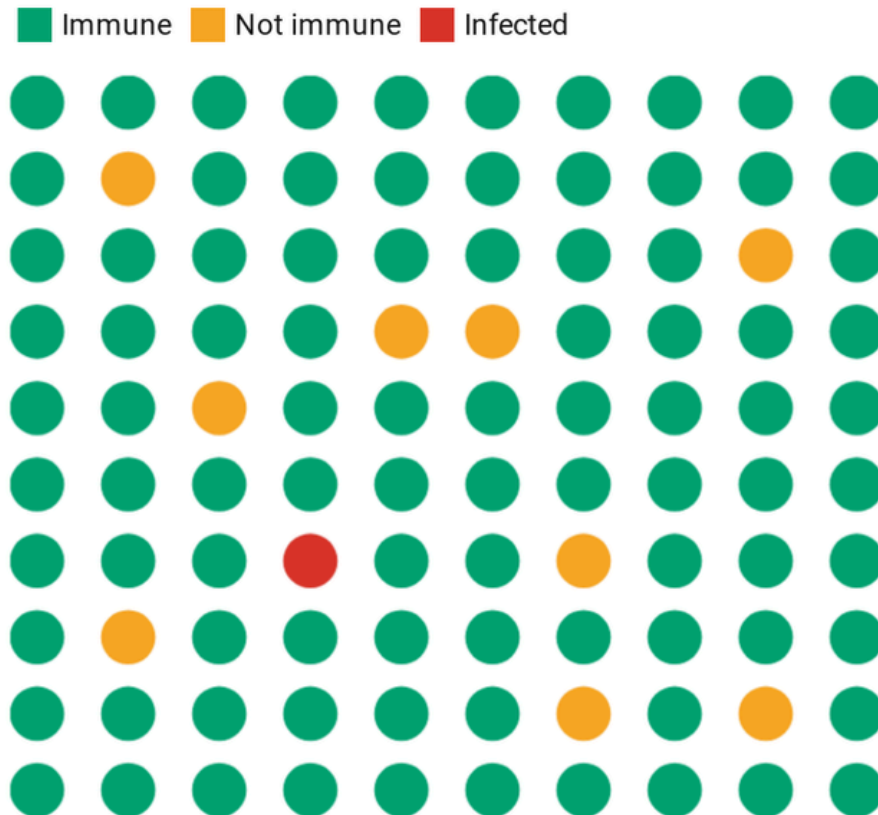
Estimating marginal damages with EZ-Pass



Positive externalities

Visualizing herd immunity

If enough people have immunity, the virus is less likely to spread because the few who aren't immune are less likely to come in contact with someone who is infected.



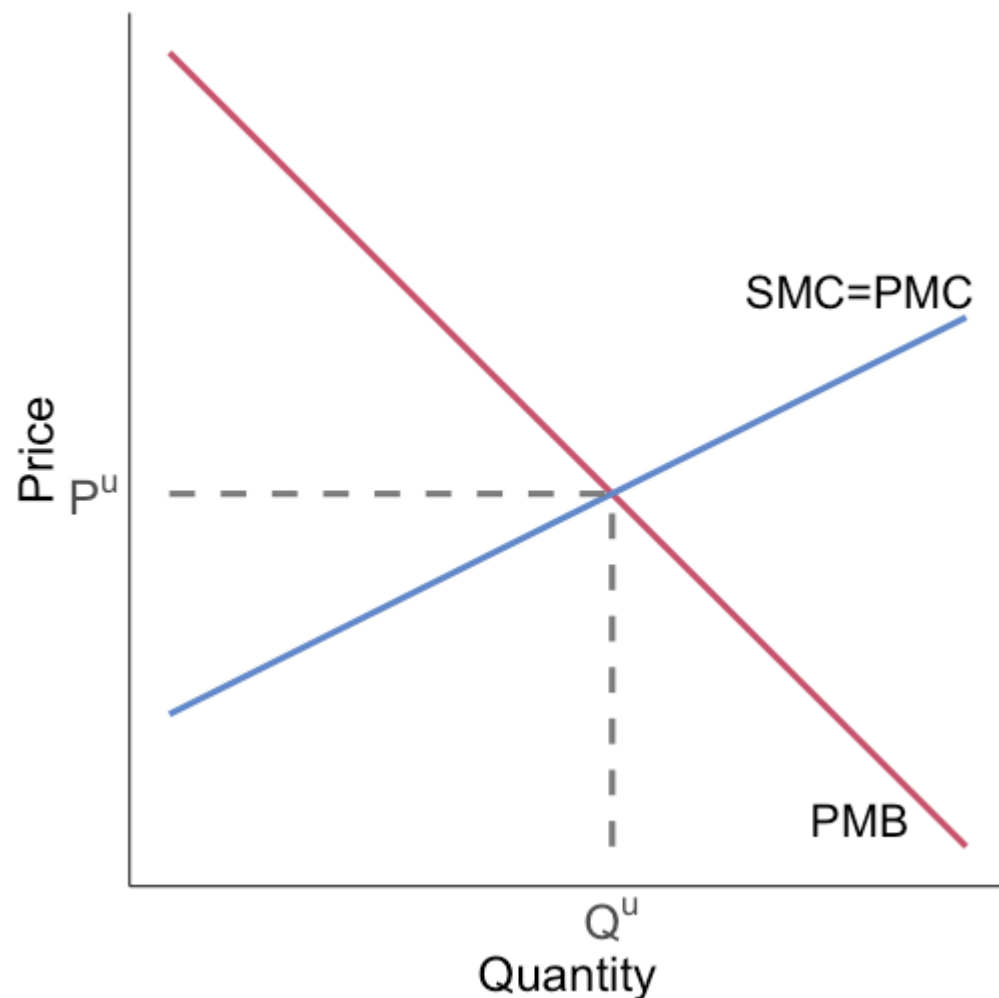
Positive externalities



Vaccines and masks are examples of goods with positive externalities

Getting a vaccine or wearing a mask can benefit people who are not involved in that decision

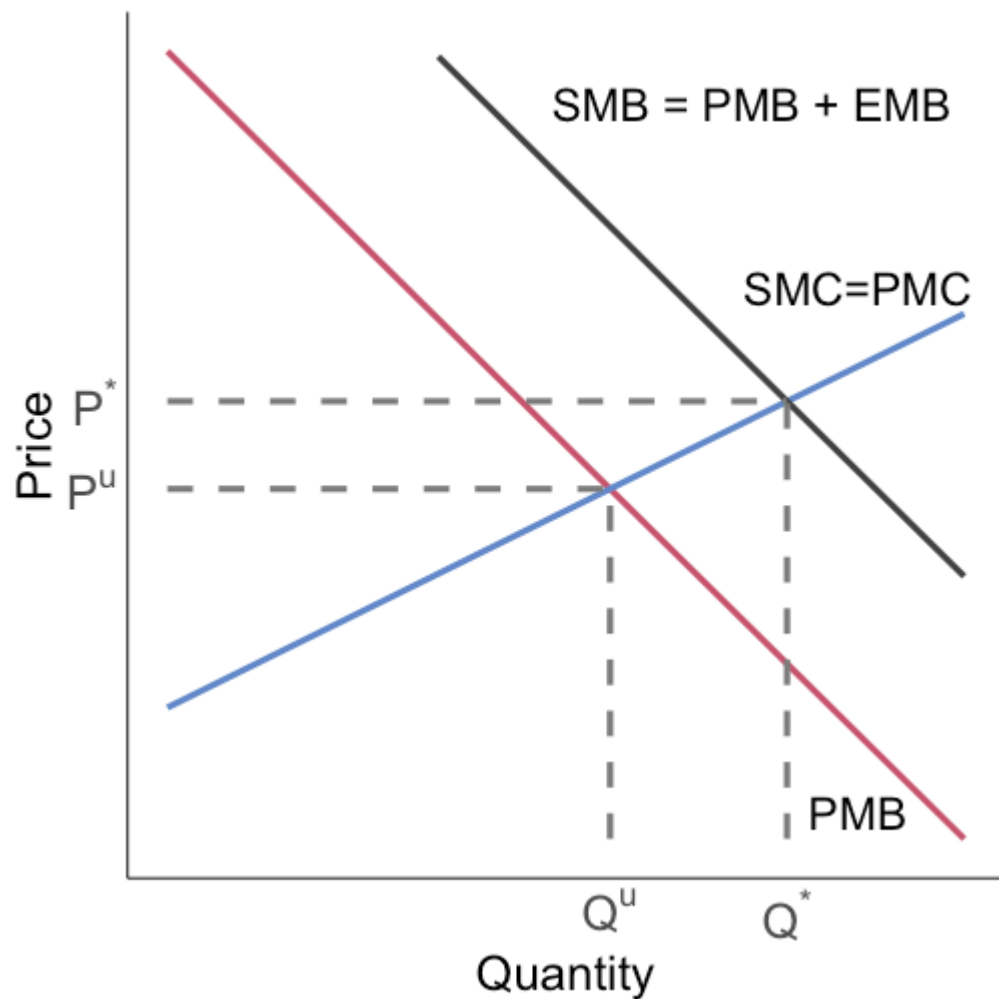
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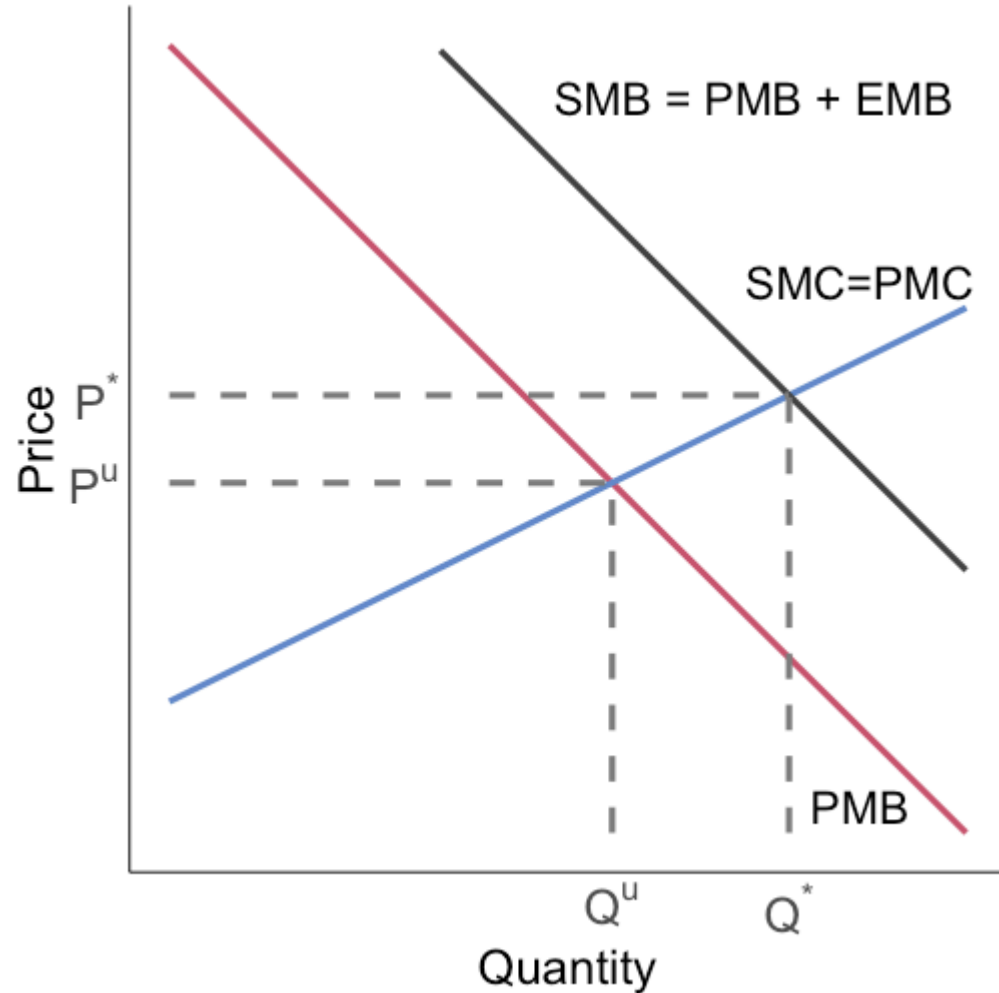
Where does the SMB curve lie?

Positive externalities: graphical



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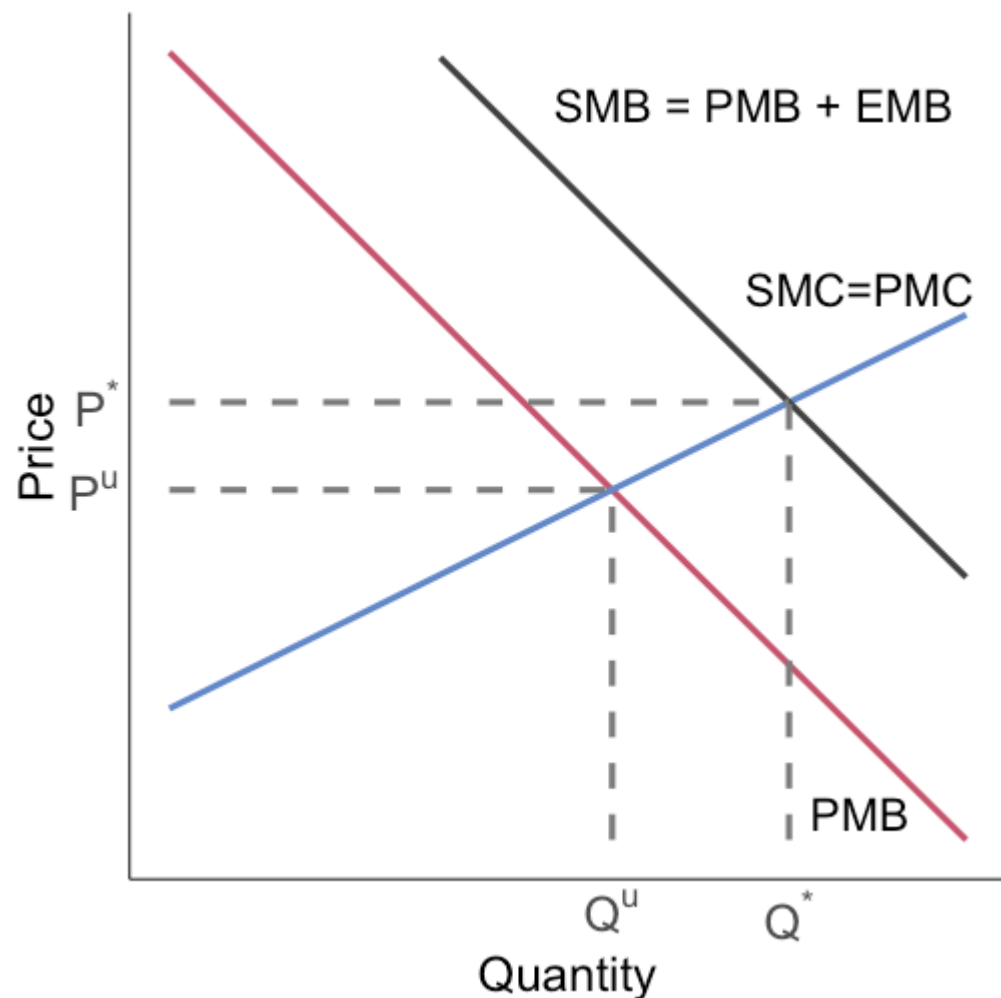
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The PMB curve only reflects the **private benefits** of getting a vaccine

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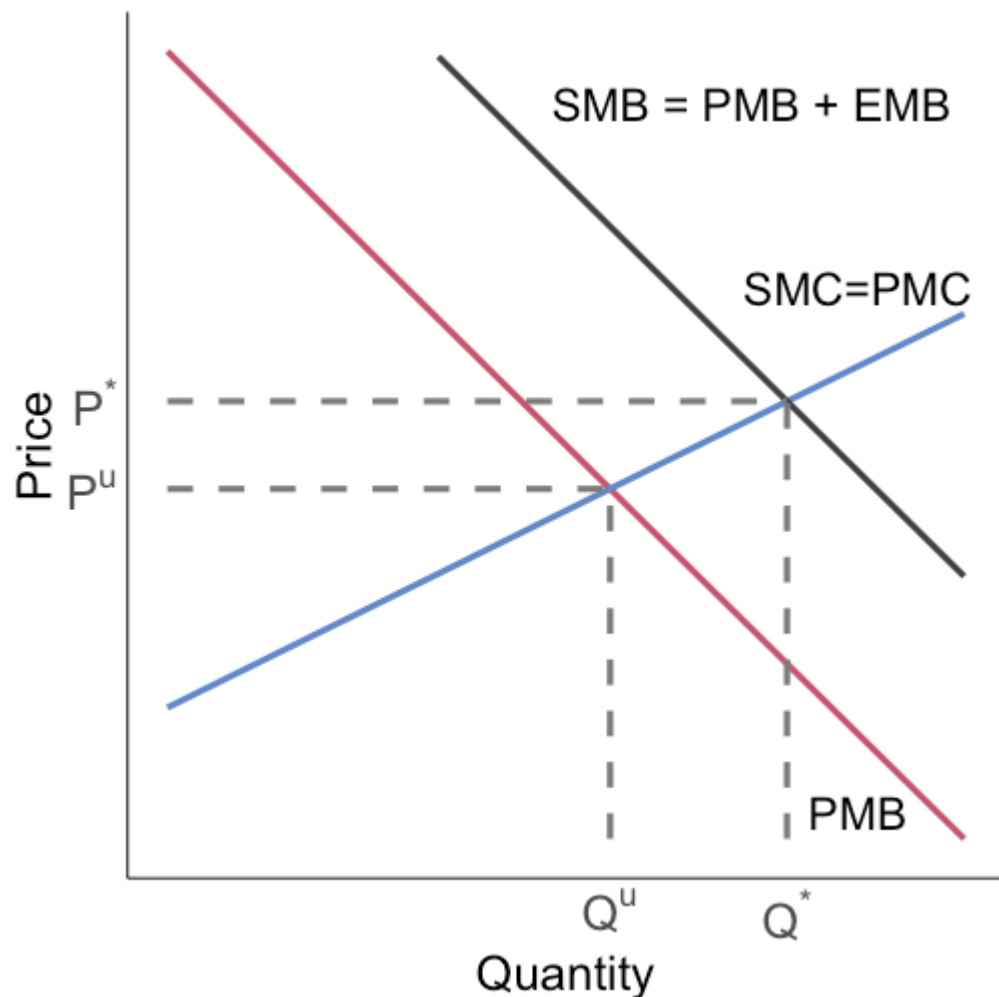


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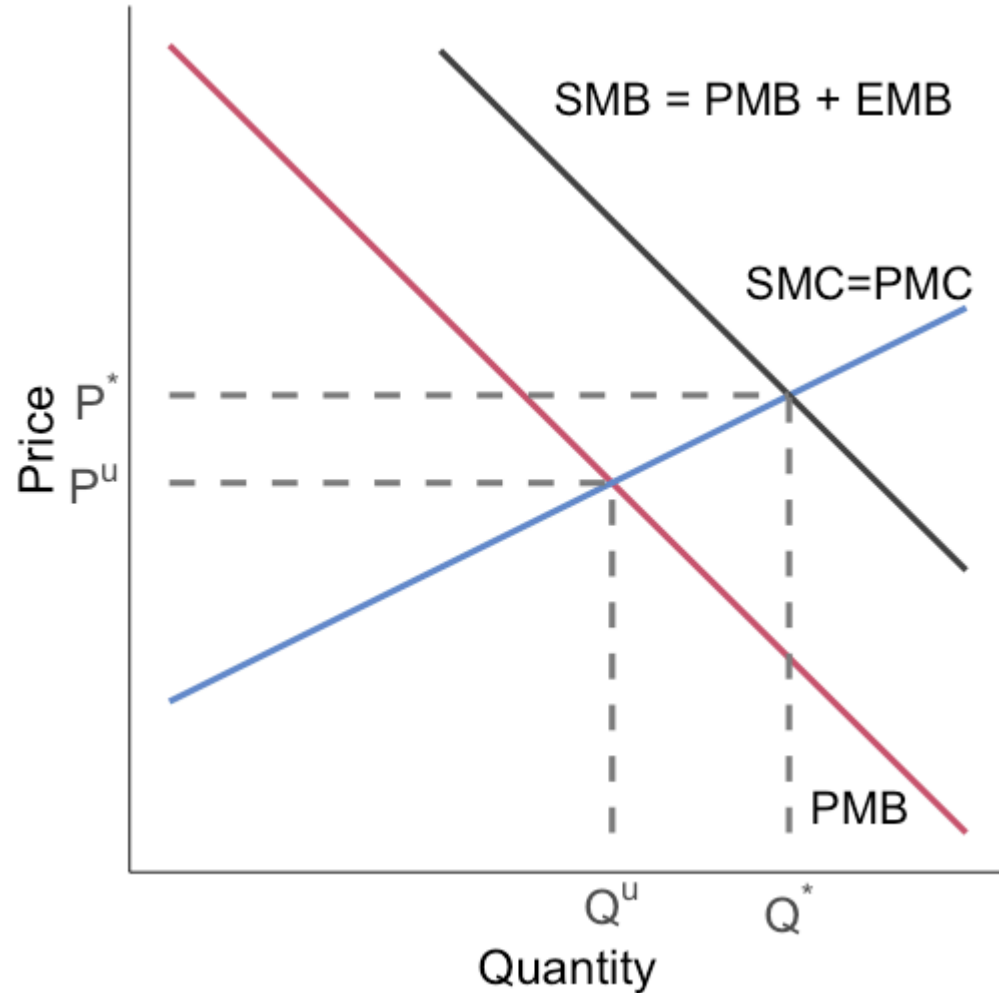
It does not account for the external herd immunity benefits

Positive externalities: graphical



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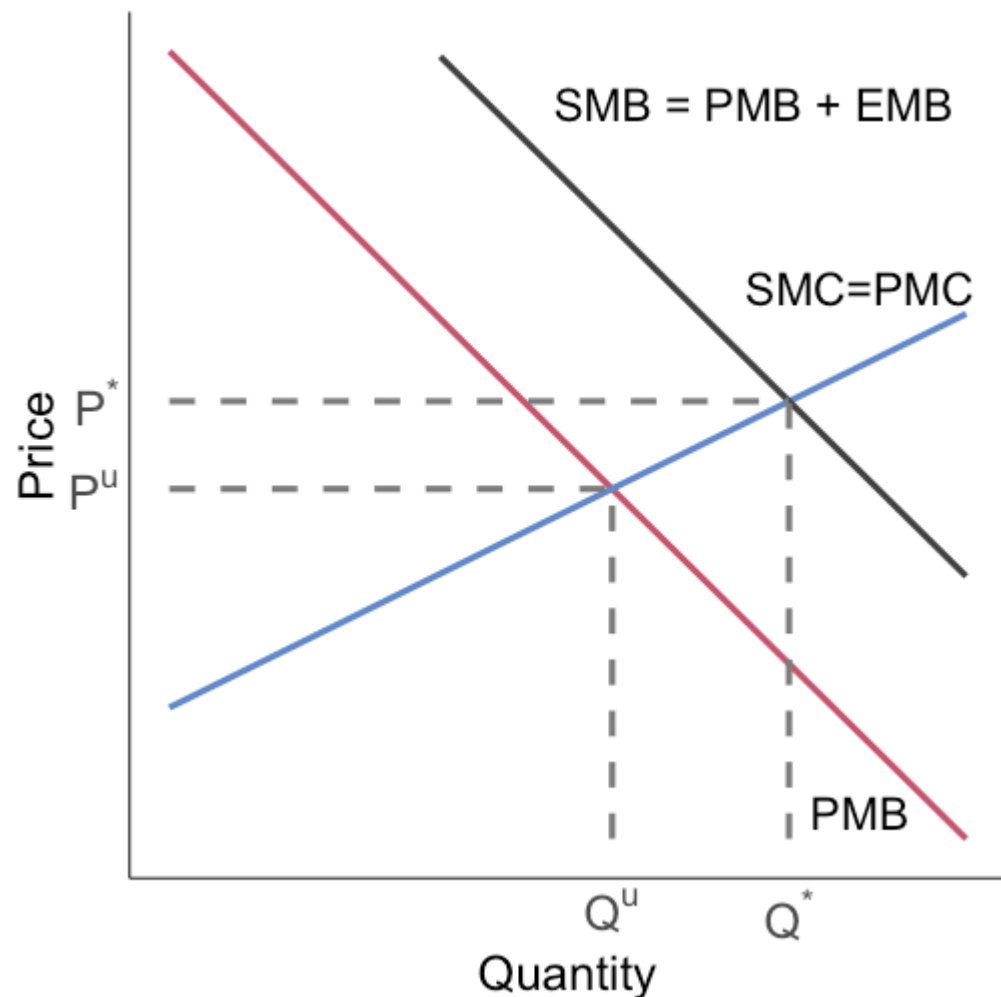
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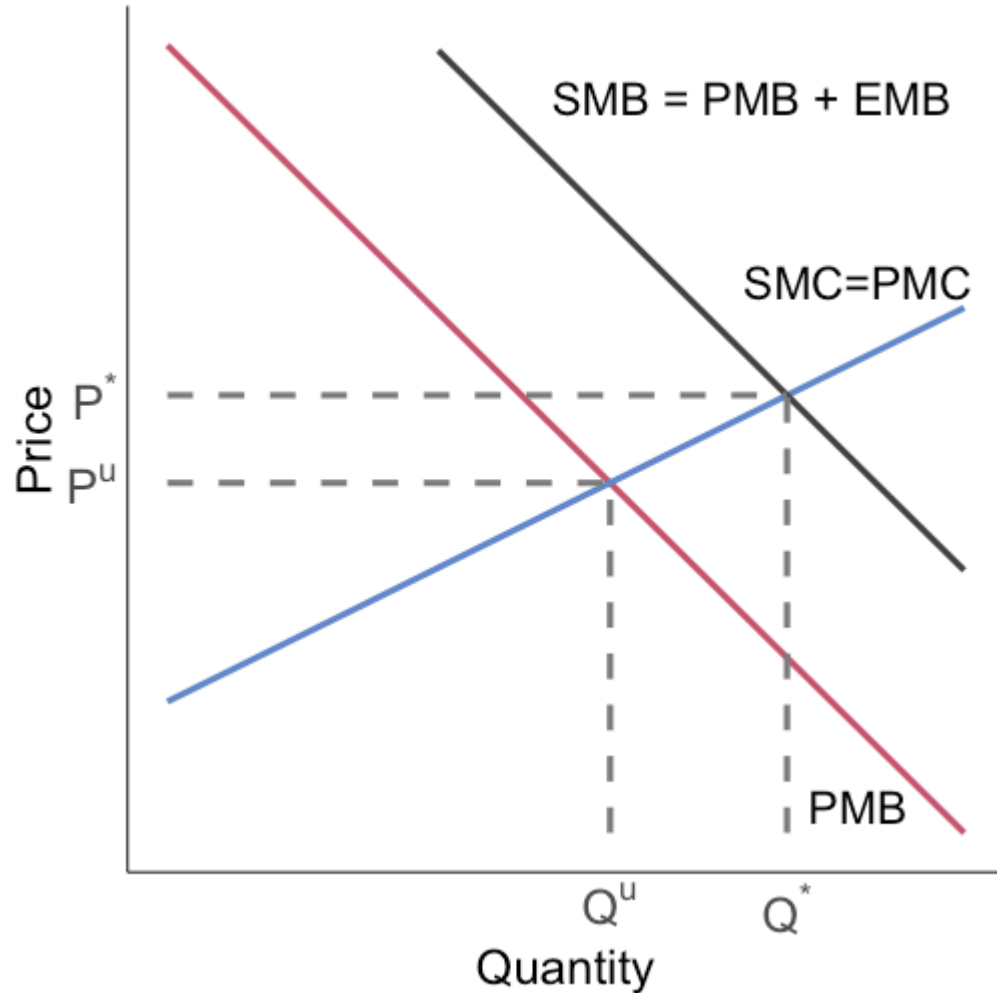


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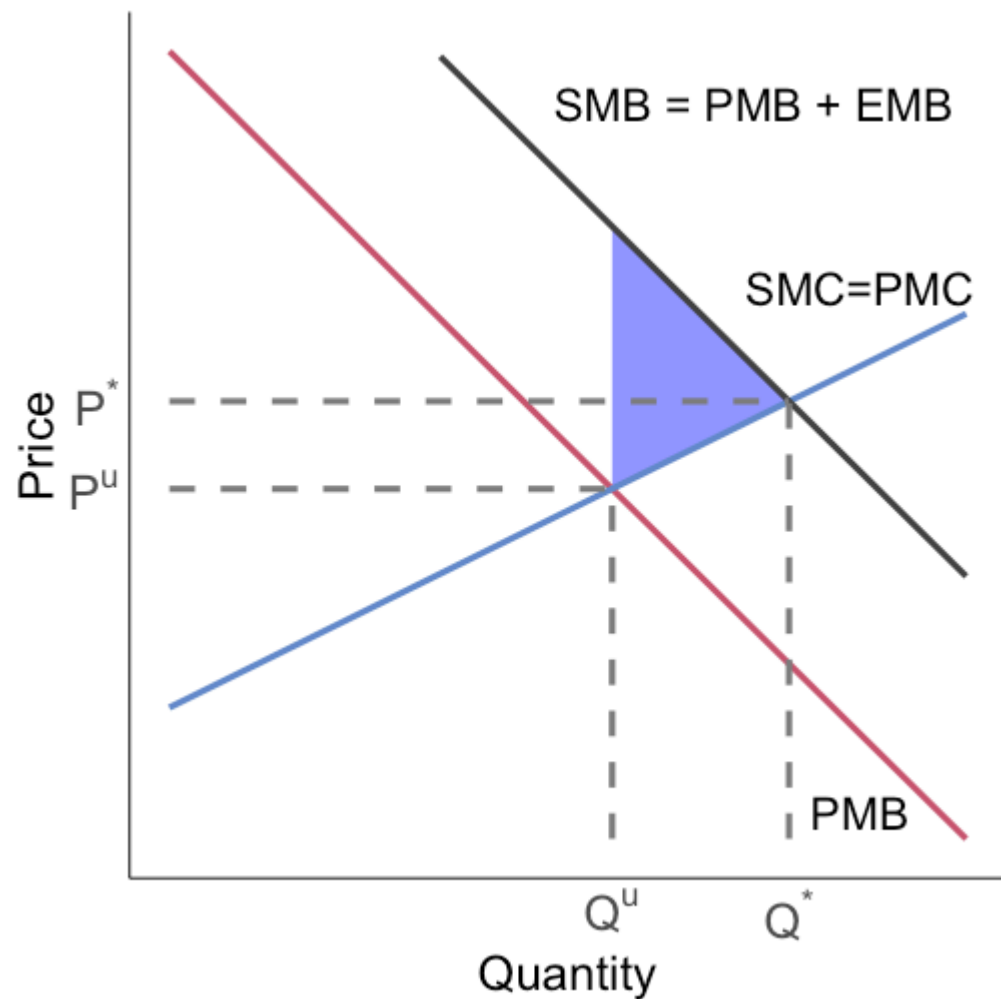
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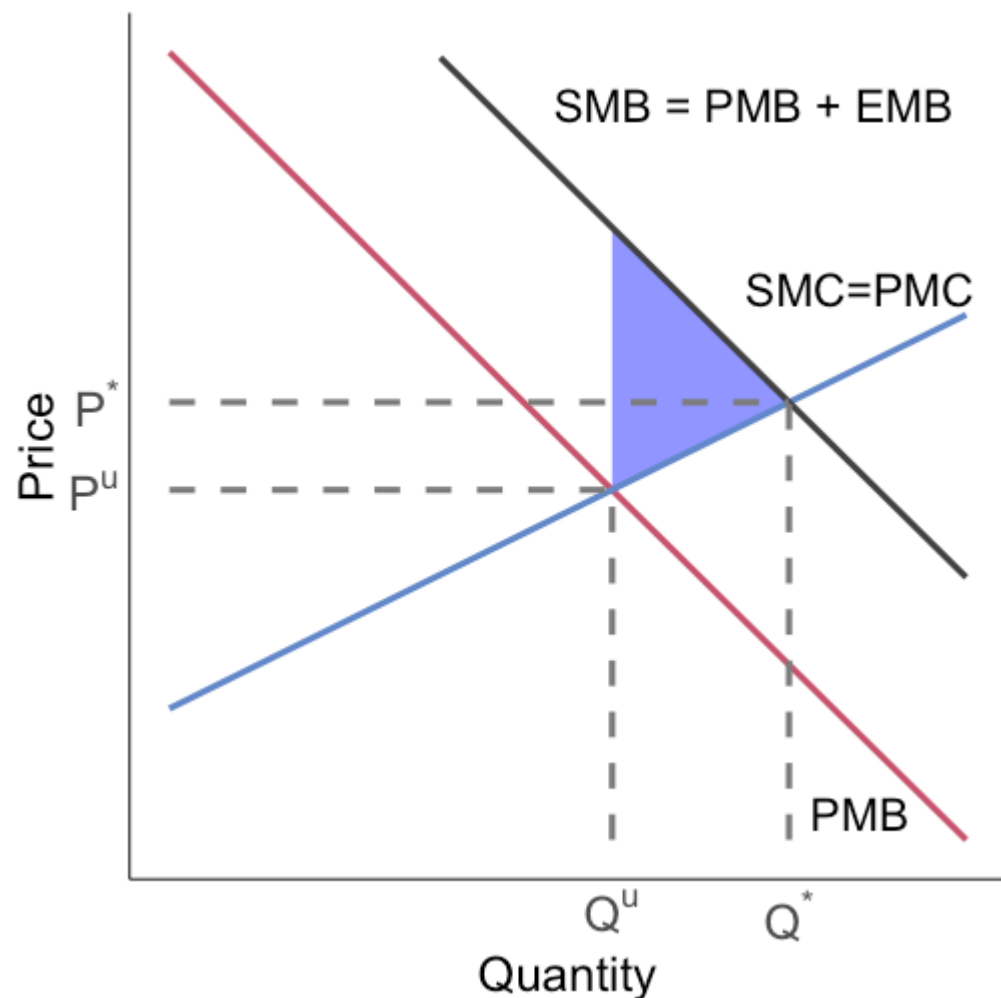
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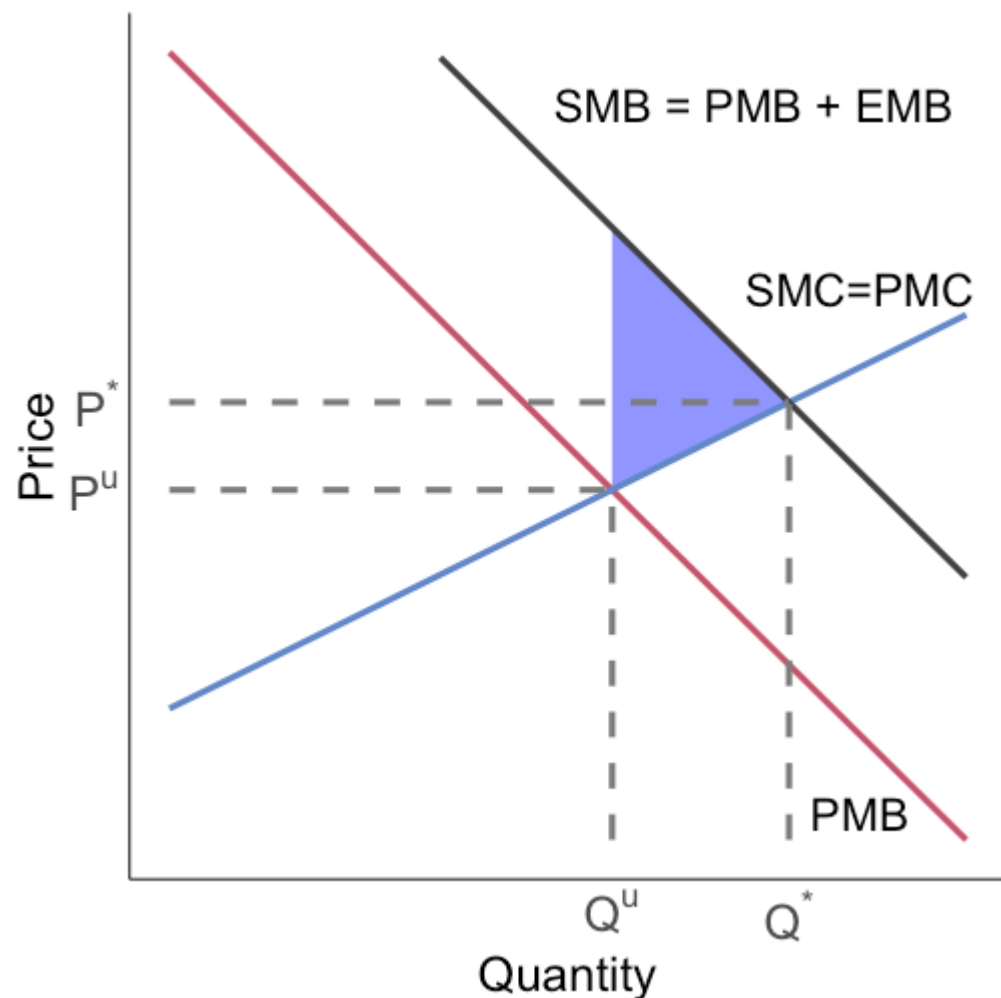


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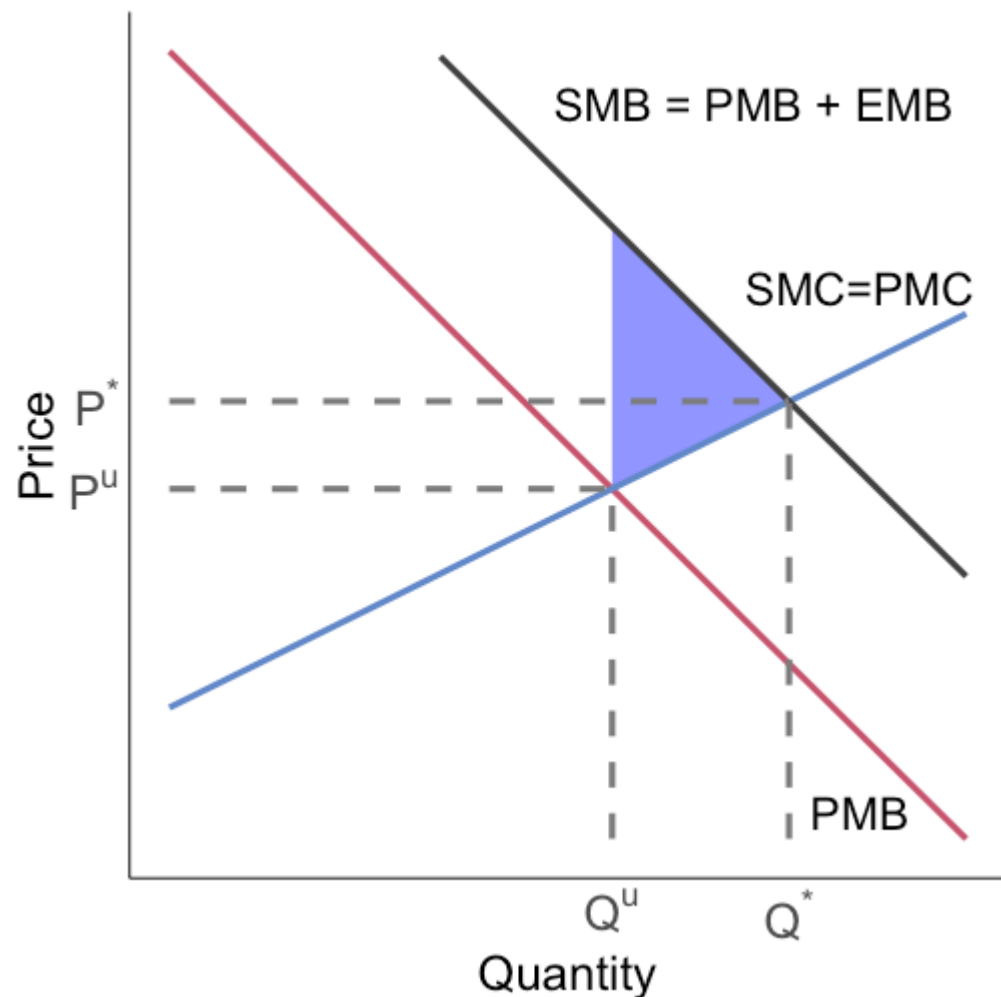


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This is the difference in SMB and SMC for units where $SMC < SMB$

This is the loss to society caused by the externality in the unregulated private market

Positive externalities: graphical



The private market produces too few vaccines

The private actors are not accounting for the external benefits they confer on people who are not in the vaccine transaction (e.g. third parties whose health is affected)

News report: the first U.S. measles deaths of 2026



Measles returned as vaccination rates fell

On August 25, 2026, Pennsylvania confirmed:

- **Two measles-associated deaths**, both among unvaccinated people
- The state's first measles-associated deaths in 35 years
- 393 confirmed cases across 28 counties during 2026

Two doses of the MMR vaccine provide approximately **97% protection**

Vaccination and infection create externalities

Vaccination can create a positive externality:

- It lowers an individual's probability of infection
- When it also lowers transmission, it protects other people

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Efficiency: private choices use PMB and PMC; social choices use SMB and SMC

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1. Poorly defined property rights
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Typically one of two reasons:

1. Poorly defined property rights

- Who owns the right to the air?

2. High transactions costs

- Hard to bargain over desired air quality with millions of people

Let's conceptualize a model of efficient bargaining using an Edgeworth Box

Why do externalities arise? Edgeworth Box

- Two individuals: A and B
- Two private goods: X and Y

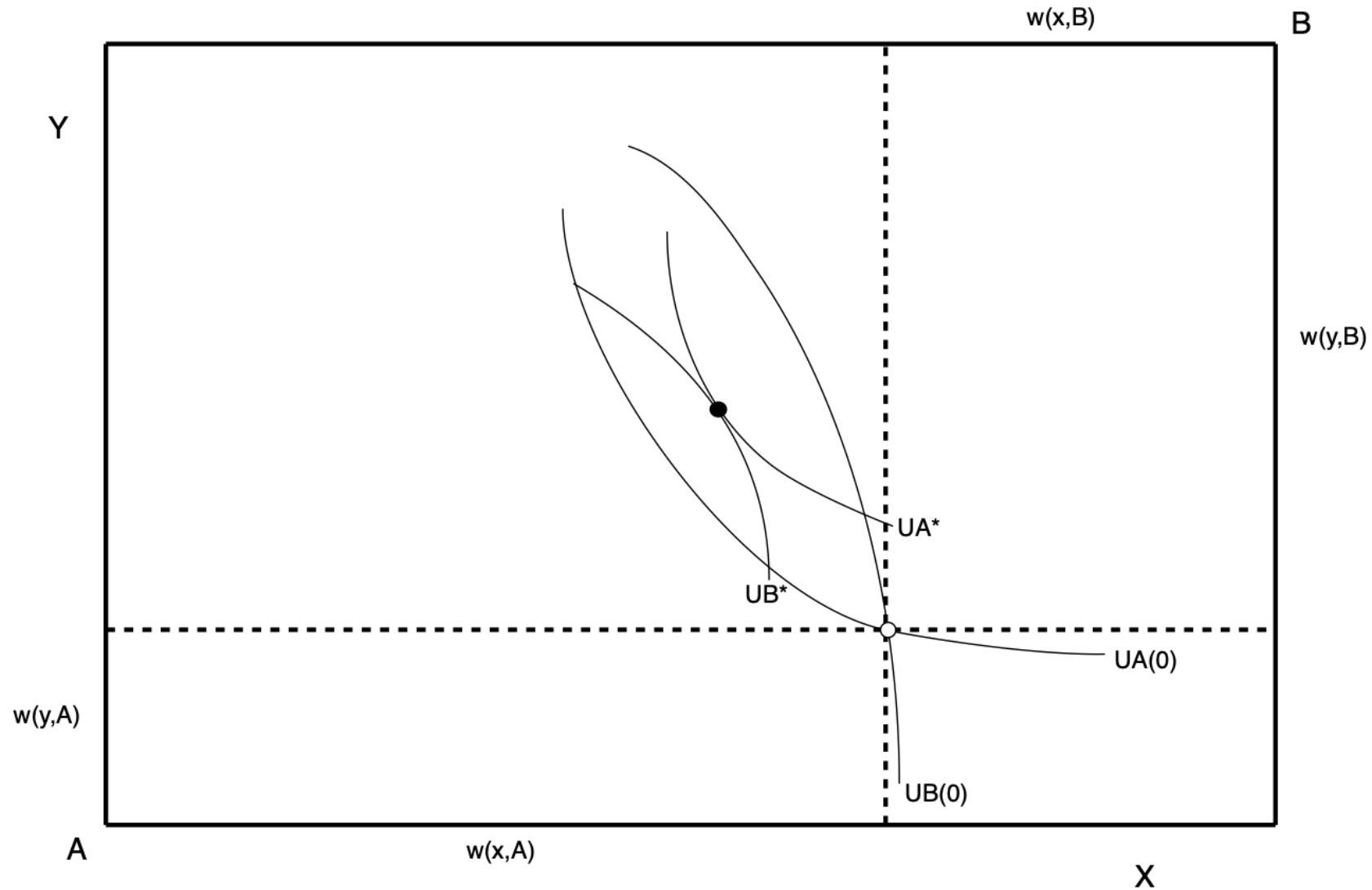
Each individual begins with an initial endowment of each good:

- $A : w_X^A, w_Y^A$
- $B : w_X^B, w_Y^B$

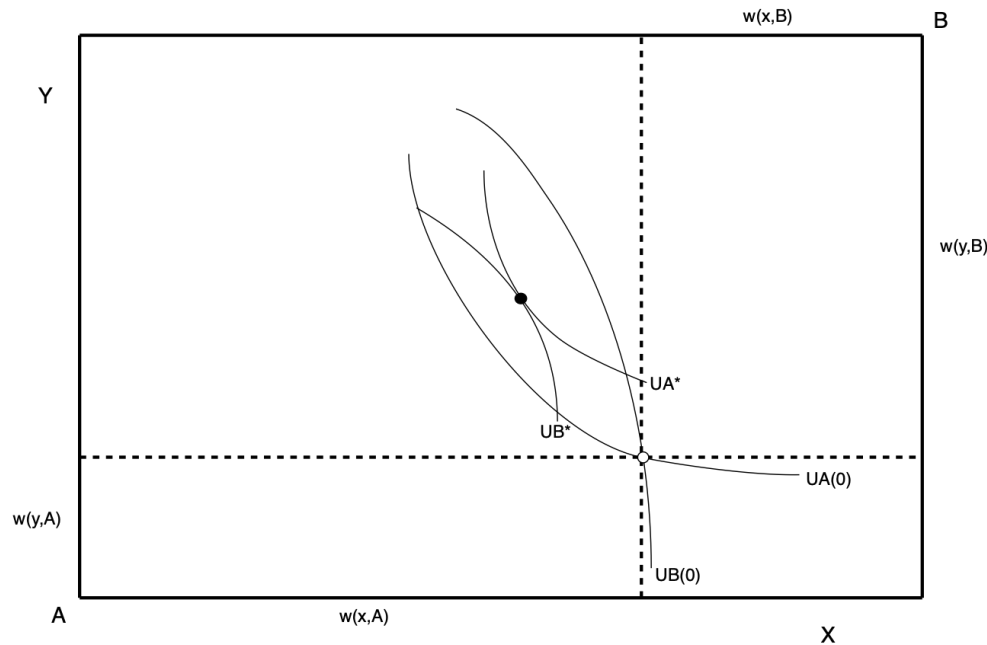
This gives us a total endowment:

- $W_X = w_X^A + w_X^B$
- $W_Y = w_Y^A + w_Y^B$

Why do externalities arise? Edgeworth Box



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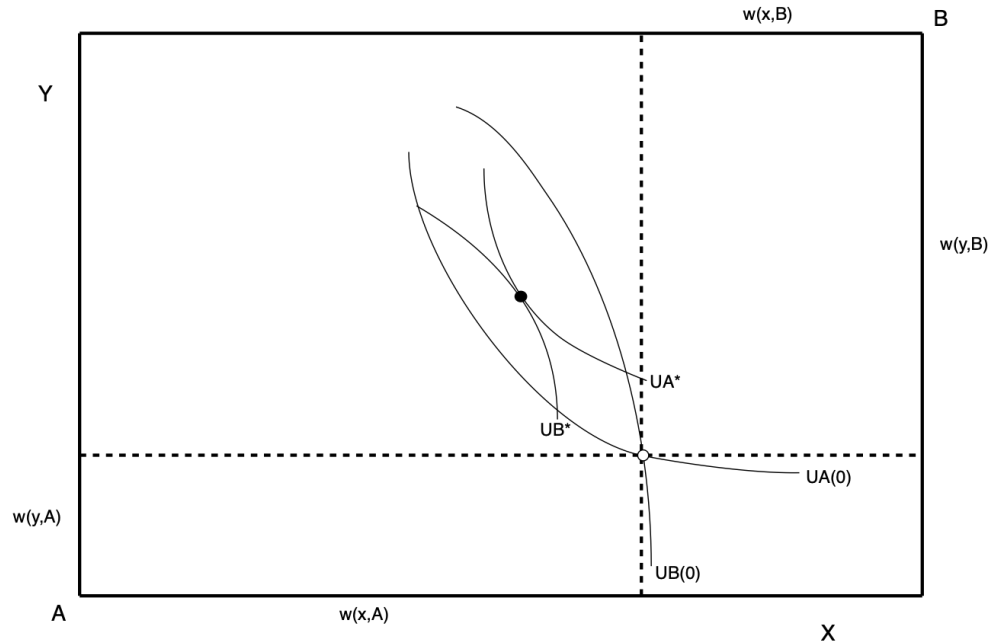
Total vertical distance is W_Y

Total horizontal distance is W_X

Initial endowment is given by the empty circle

Initial indifference curves for A and B are $UA(0)$ and $UB(0)$

Why do externalities arise? Edgeworth Box



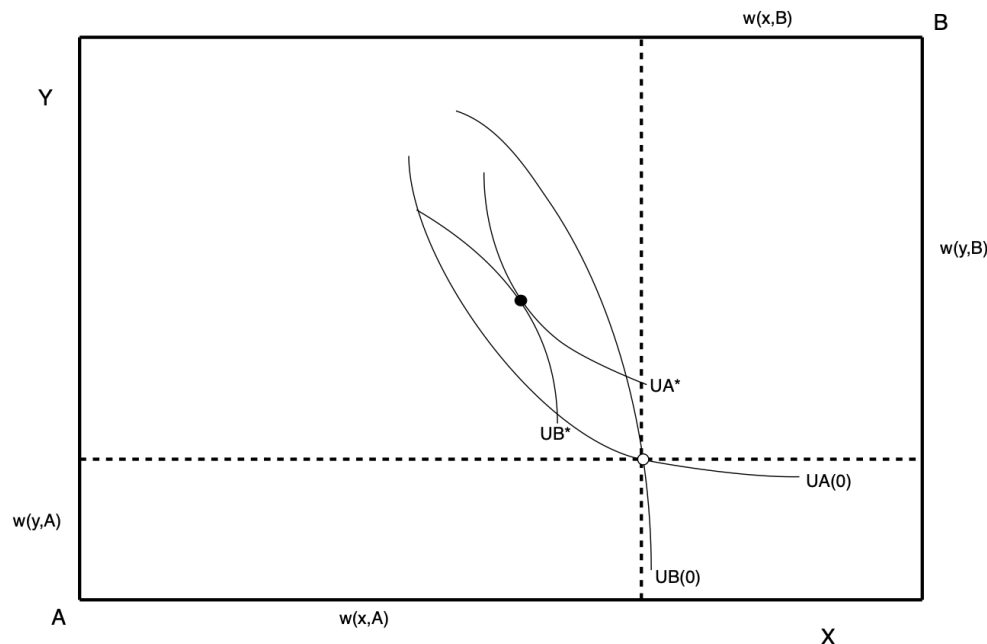
Is there a possible Pareto improvement?

e.g. can we make both A and B better off?

Yes!



Why do externalities arise? Edgeworth Box



Yes!

If we move anywhere in the lens of their initial indifference curves we have a Pareto improvement

If we move to an allocation where their indifference curves are **tangent** to one another (e.g. the filled-in point), we have a Pareto optimum

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- The endowment point is well-established

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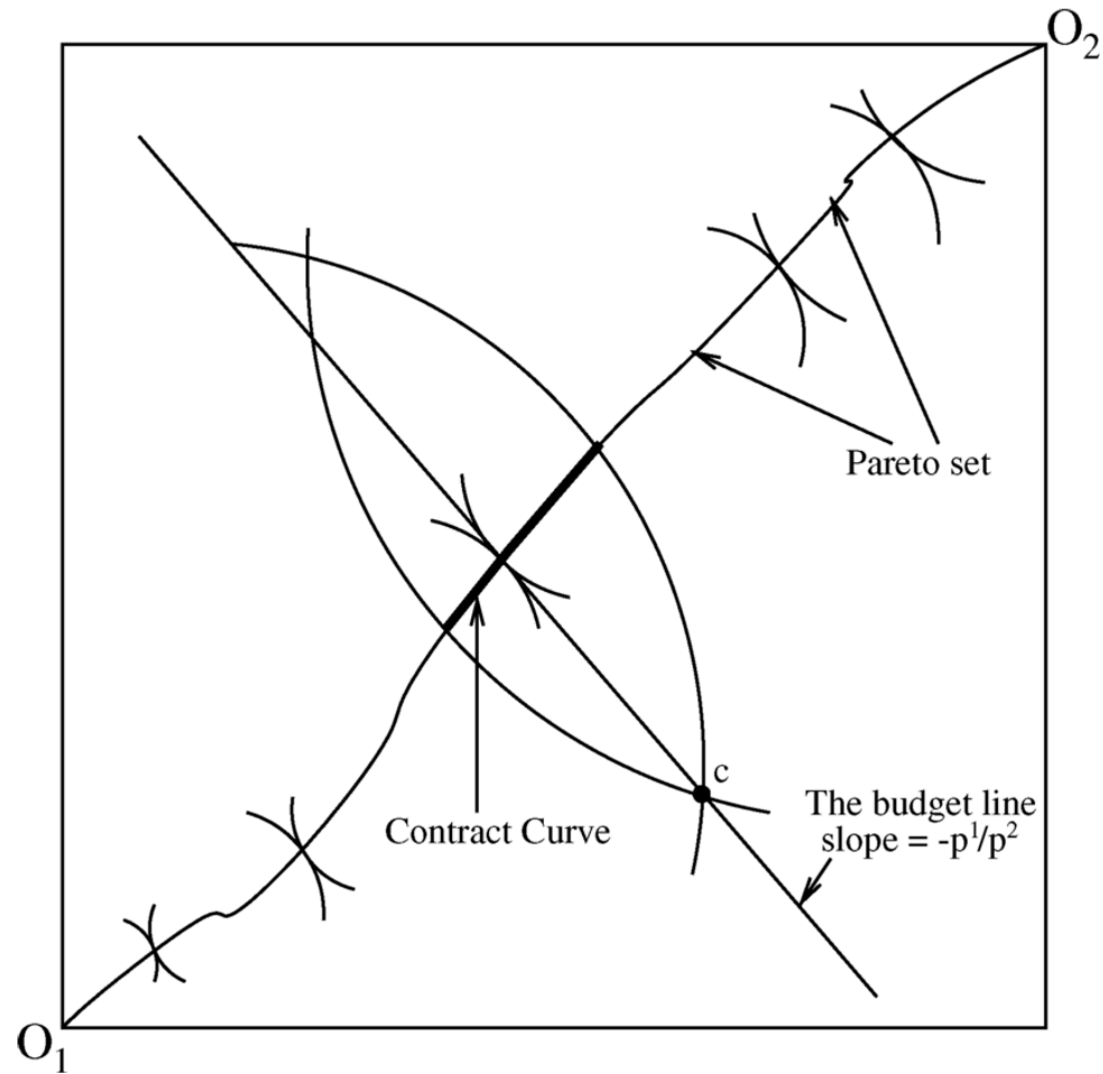
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Why do externalities arise? Edgeworth Box

In a properly functioning market:

- The endowment point is well-established
- A and B can trade X and Y to some Pareto improving point
- They continue trading until they achieve a Pareto optimal allocation
- This allocation lies on the **contract curve**: the line consisting of all Pareto efficient allocations

Why do externalities arise? Edgeworth Box



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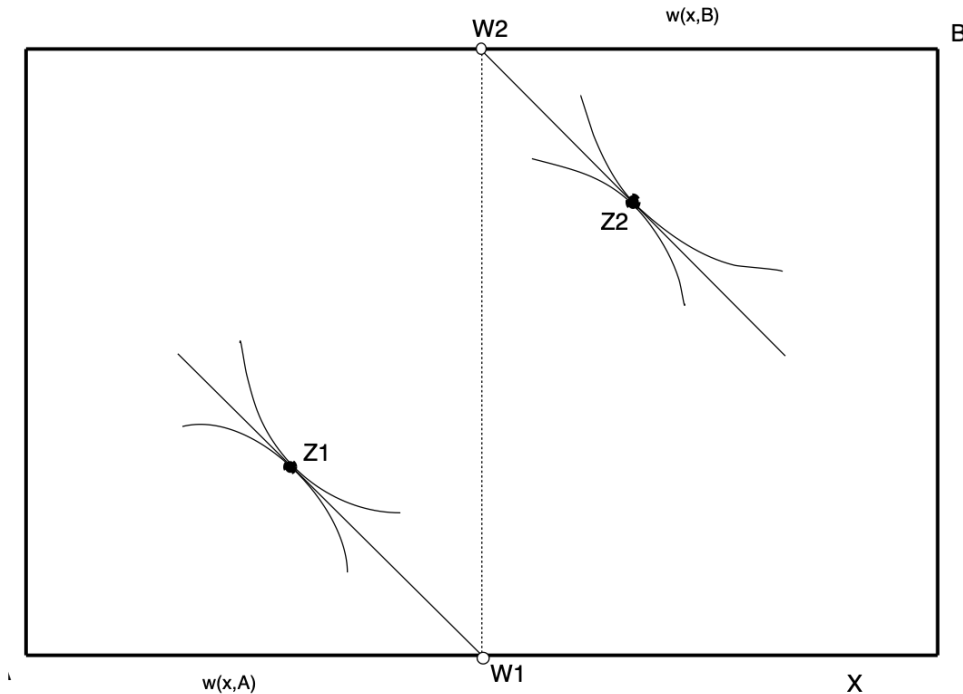
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Unlike our regular Edgeworth Box, now Y increases for **both** A and B as we move to the top of the slide (before Y increased for B as we moved to the bottom)

Suppose that A likes Y, but B does not

Suppose both start off with the same quantity of X

Why do externalities arise? Edgeworth Box

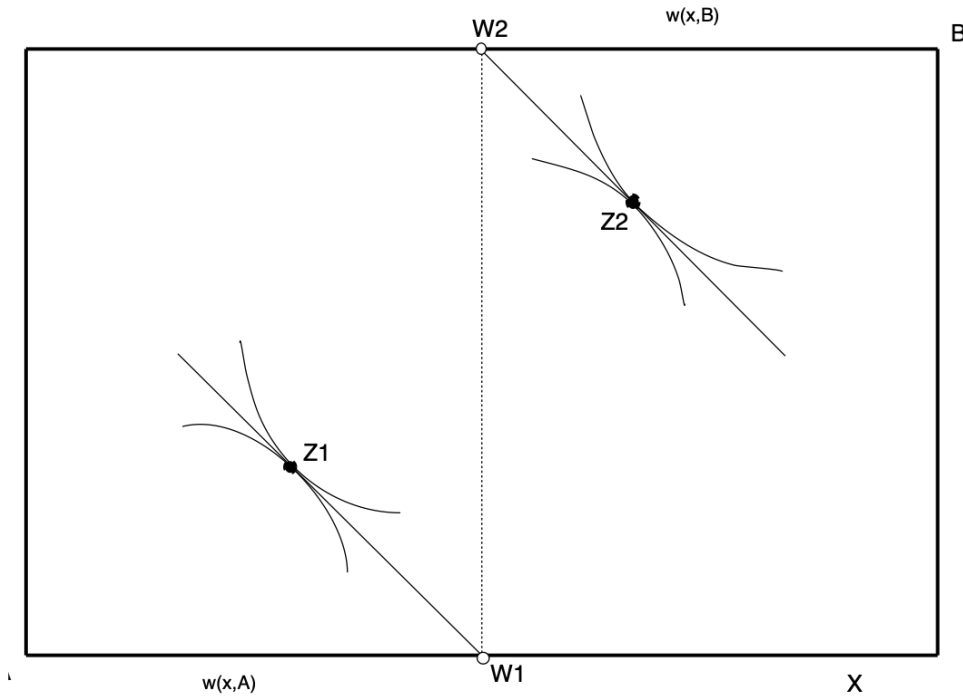


Depending on who has property rights, we either start at:

- $W1$ (B has property rights)
- $W2$ (A has property rights)

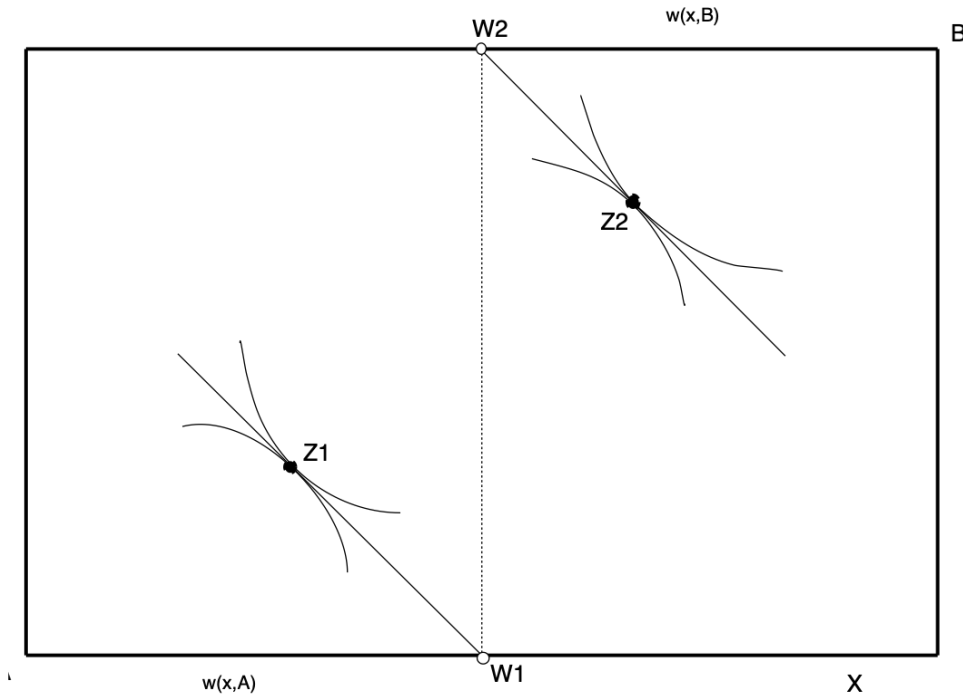
Think about why these are where we must start

Why do externalities arise? Edgeworth Box



Suppose we start at $W1$, what happens?

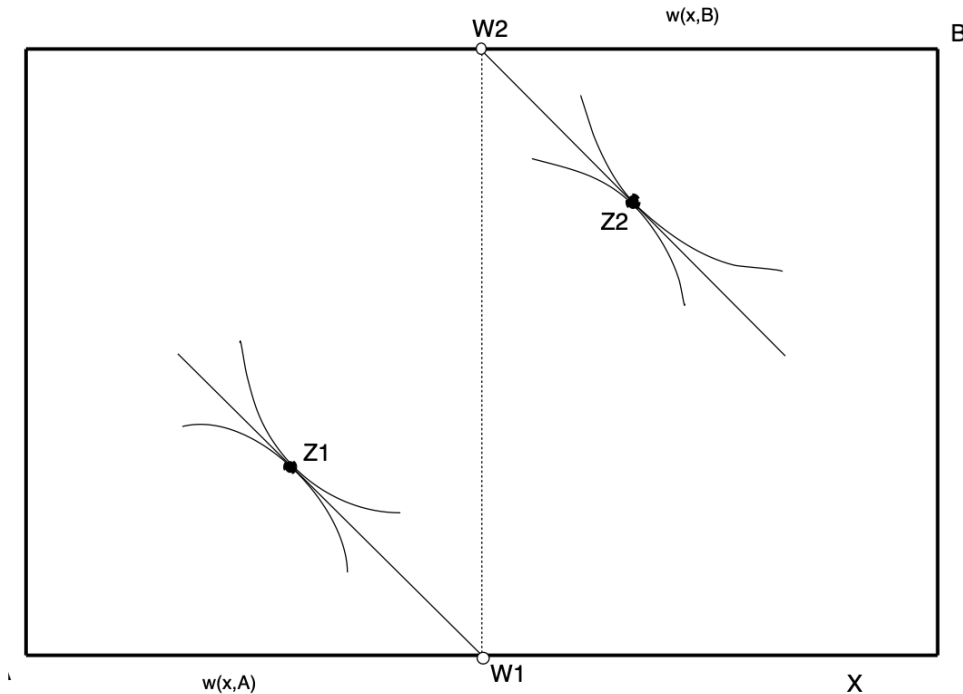
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Suppose we start at $W1$, what happens?

A wants to have more Y, but this imposes a cost on B

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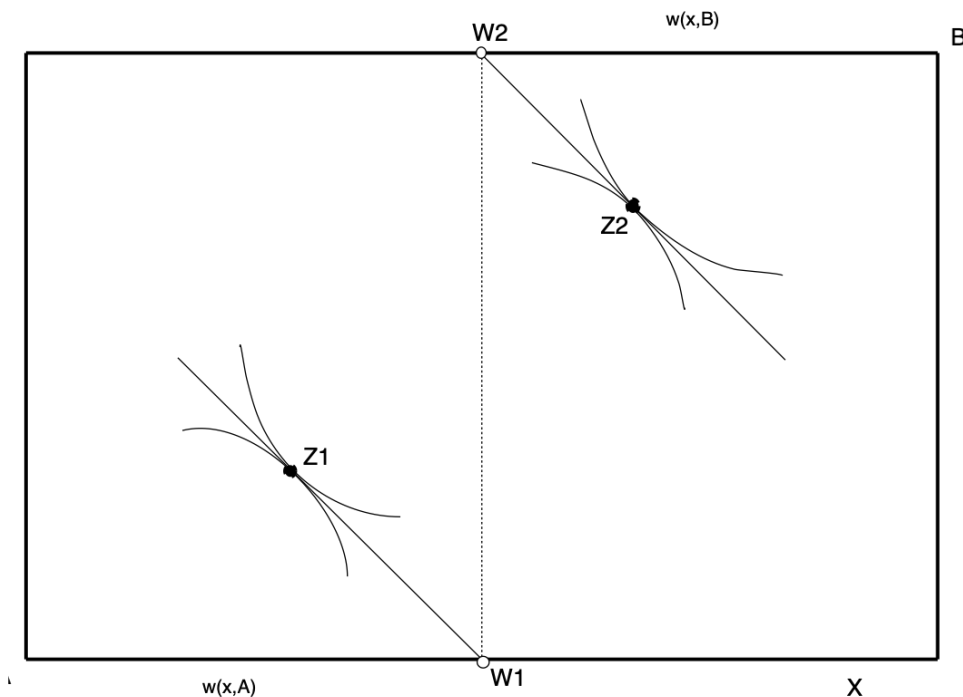


Suppose we start at $W1$, what happens?

A wants to have more Y , but this imposes a cost on B

Therefore, A has to **pay** B to get more Y

Why do externalities arise? Edgeworth Box



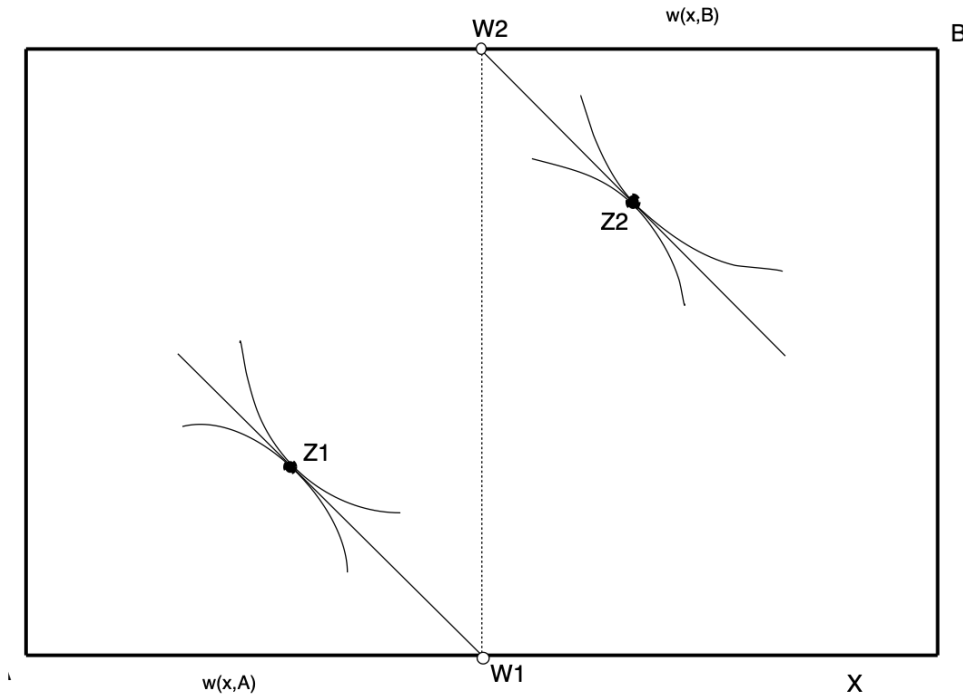
Suppose we start at W_1 , what happens?

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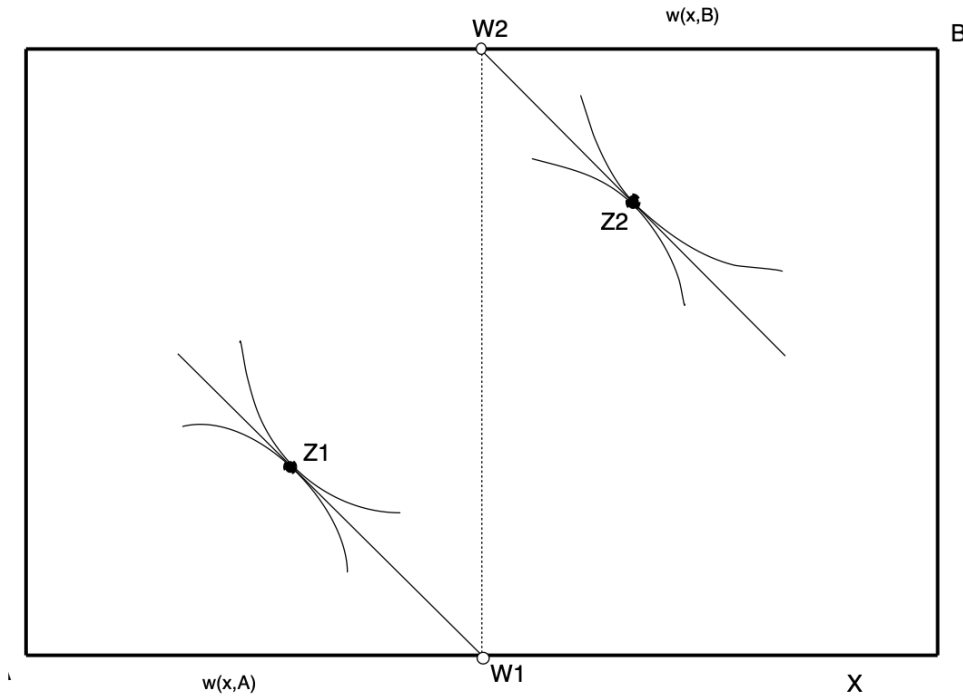
A pays B in units of X, move to Z_1 , Pareto optimum

Why do externalities arise? Edgeworth Box



Suppose we start at $W2$, what happens?

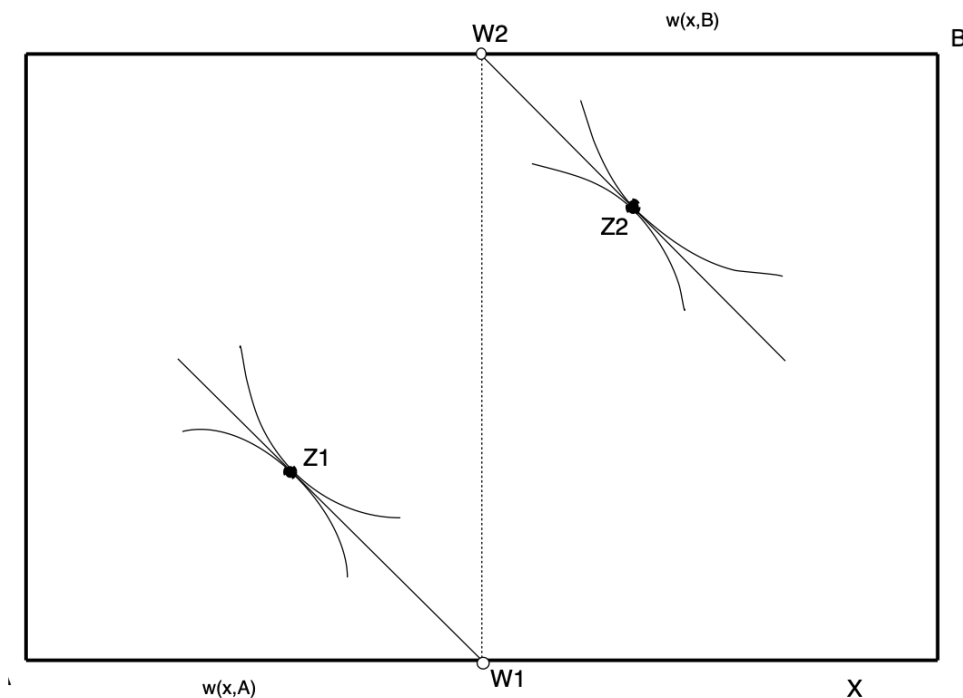
Why do externalities arise? Edgeworth Box



Suppose we start at $W2$, what happens?

B wants to have less Y , but this imposes a cost on A

Why do externalities arise? Edgeworth Box

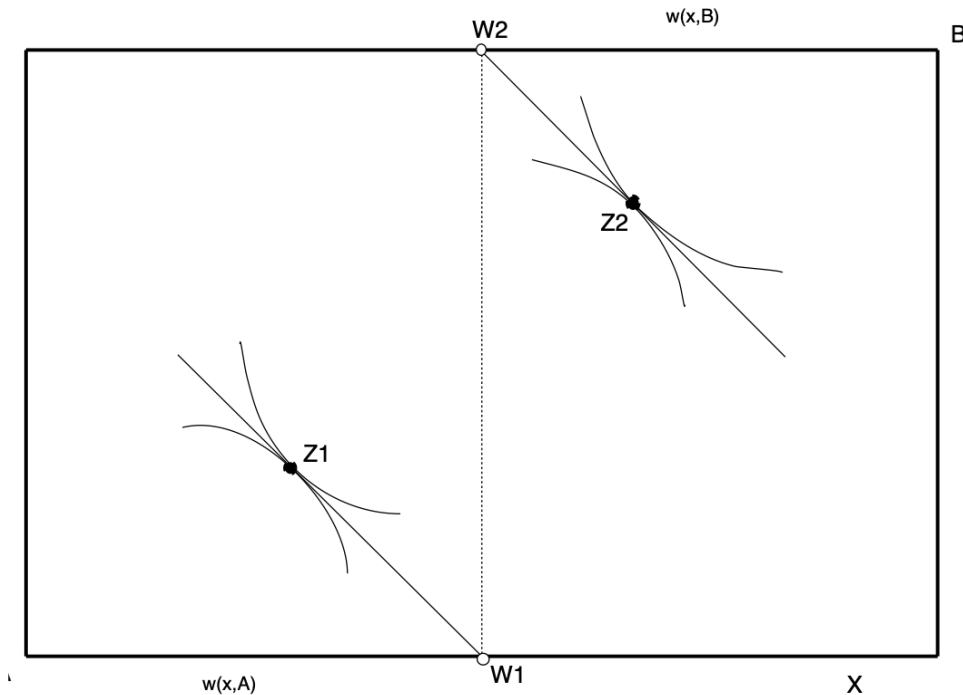


Suppose we start at $W2$, what happens?

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Therefore, B has to **pay** A to get less Y

Why do externalities arise? Edgeworth Box



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In the previous example we were able to achieve the Pareto optimum even with a public good / externality

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Why?

Why do externalities arise? Edgeworth Box

In the previous example we were able to achieve the Pareto optimum even with a public good / externality

Why?

1. Property rights were assigned to either A or B
2. Transactions costs were low (didn't have to pay a fee to trade X)

Property rights and externalities

A solution to many externalities is to just assign property rights and let the market do its thing

Property rights and externalities

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We'll talk about a few ways that we can assign property rights

Transactions costs and externalities

Now suppose there were many non-smokers

Transactions costs and externalities

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Even if they were assigned the property rights, it might be hard for them to bargain

- Takes a lot of time to find something that works for everyone
- Negotiating over how much X each person gets

Transactions costs and externalities

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Even if they were assigned the property rights, it might be hard for them to bargain

- Takes a lot of time to find something that works for everyone
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The costs of bargaining may exceed the benefits and we end up stuck at W_2

Transactions costs and externalities

Road noise: drivers implicitly have property rights to make noise around roads

Transactions costs and externalities

Road noise: drivers implicitly have property rights to make noise around roads

Even if you prefer quiet, you cannot negotiate a payment with every loud car that might pass

The free-rider problem

Externalities and public goods/bads often exhibit many of the same features

Both are subject to the **Free-Rider Problem**

A type of market failure that occurs when those who benefit from resources, public goods (such as public roads or hospitals), or services of a communal nature do not pay for them or under-pay

e.g.

- people don't pay their taxes for publicly-provided services
- non-smokers will wait for others to pay in order to reduce smoke

The provision of public goods

Public goods

How do we efficiently provide public goods?

We know:

- Private goods: $PMB = PMC \leftrightarrow SMB = SMC$
- Goods with negative externalities: $PMB = SMC \leftrightarrow SMB = SMC$
- goods with positive externalities: $SMB = PMC \leftrightarrow SMB = SMC$

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- goods with positive externalities: $SMB = PMC \Leftrightarrow SMB = SMC$

Suppose we have a public good, e.g. depth of a river for public use

How do we decide the socially efficient depth?

Public goods

Optimal provision is **always** given by: $SMB = SMC$

What are the SMB and SMC for a public good?

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Different roles: non-rivalry shapes aggregation; non-excludability causes free riding

Optimal provision of public goods

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e.g. the free market underprovides clean air, national defense, etc

Modeling the provision of public goods

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Private goods: we add demand curves horizontally

Modeling the provision of public goods

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Modeling the provision of public goods

For public goods:

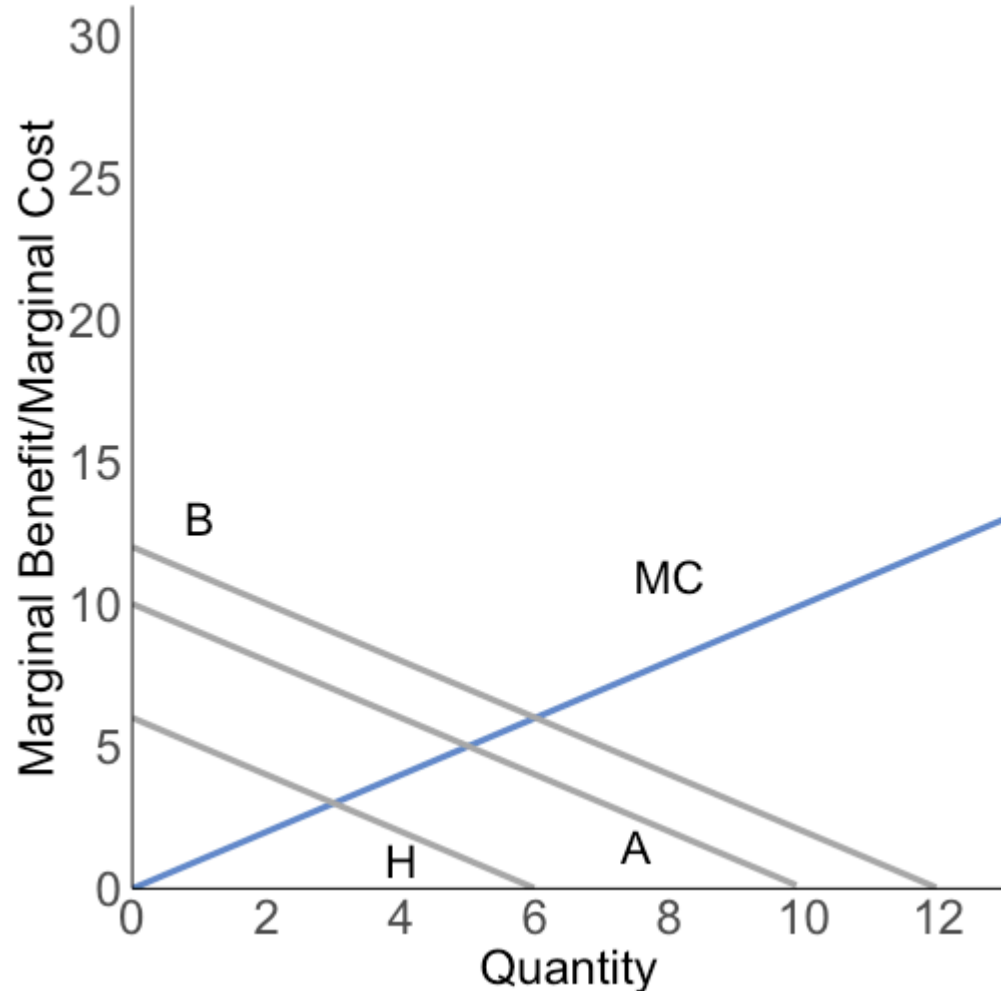
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At each quantity, what is the total marginal benefit?

At each quantity, we need to add up PMBs/prices

Public goods: we add demand curves vertically

Public goods: graphical



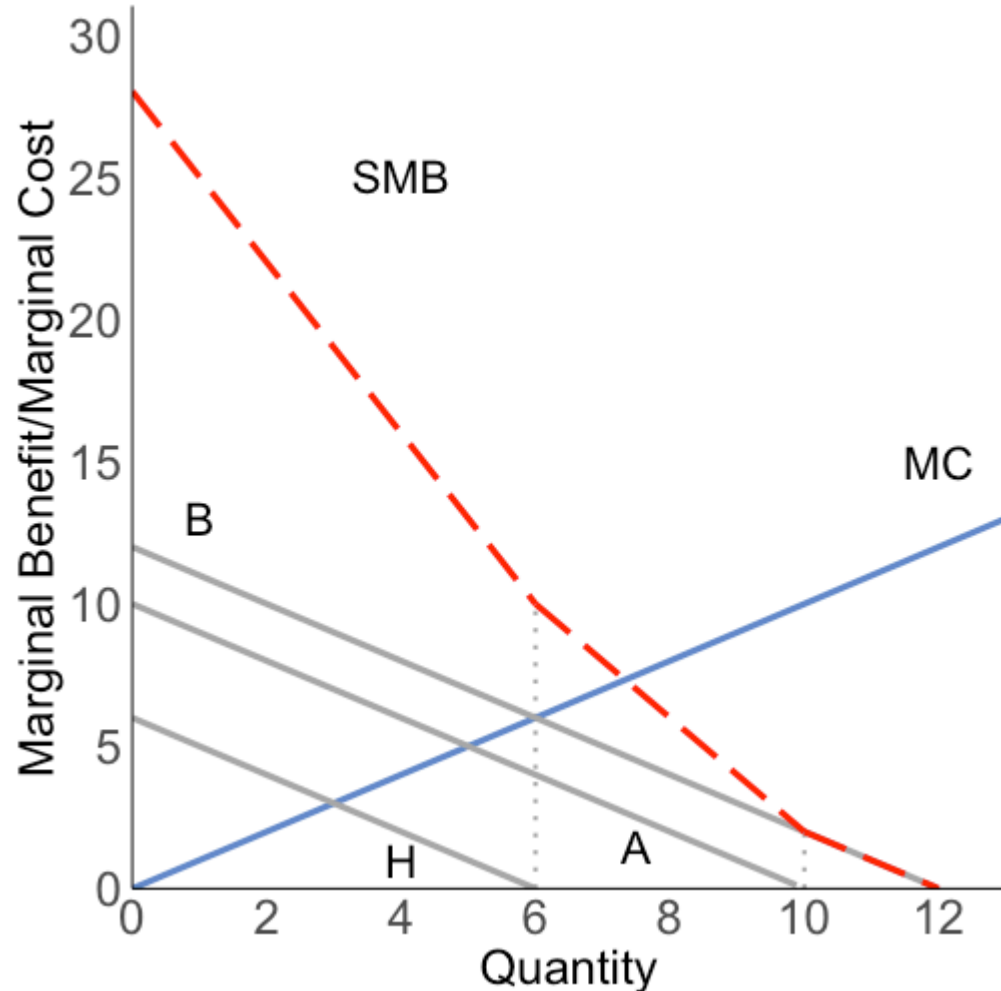
3 different groups: boaters (B), anglers (A), and hikers (H)

Each has a different marginal benefit for water depth:

- Boaters: $MB = 12 - Q$
- Anglers: $MB = 10 - Q$
- Hikers: $MB = 6 - Q$
- MC of provision: $MC = Q$

We assume MB can't go negative

Public goods: graphical

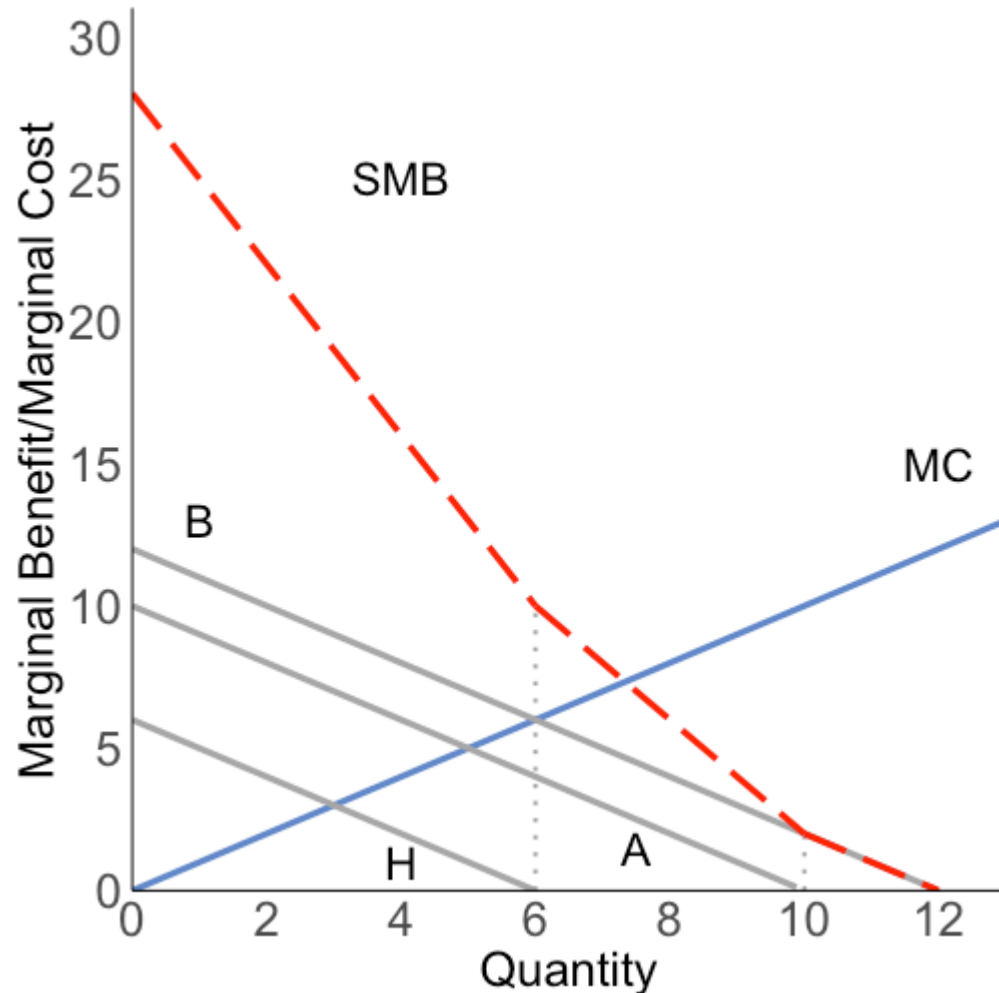


Now we need to aggregate them to get the **social marginal benefit**

We do so by adding up the demand curves vertically:

At each Q, sum the MBs

Public goods: graphical

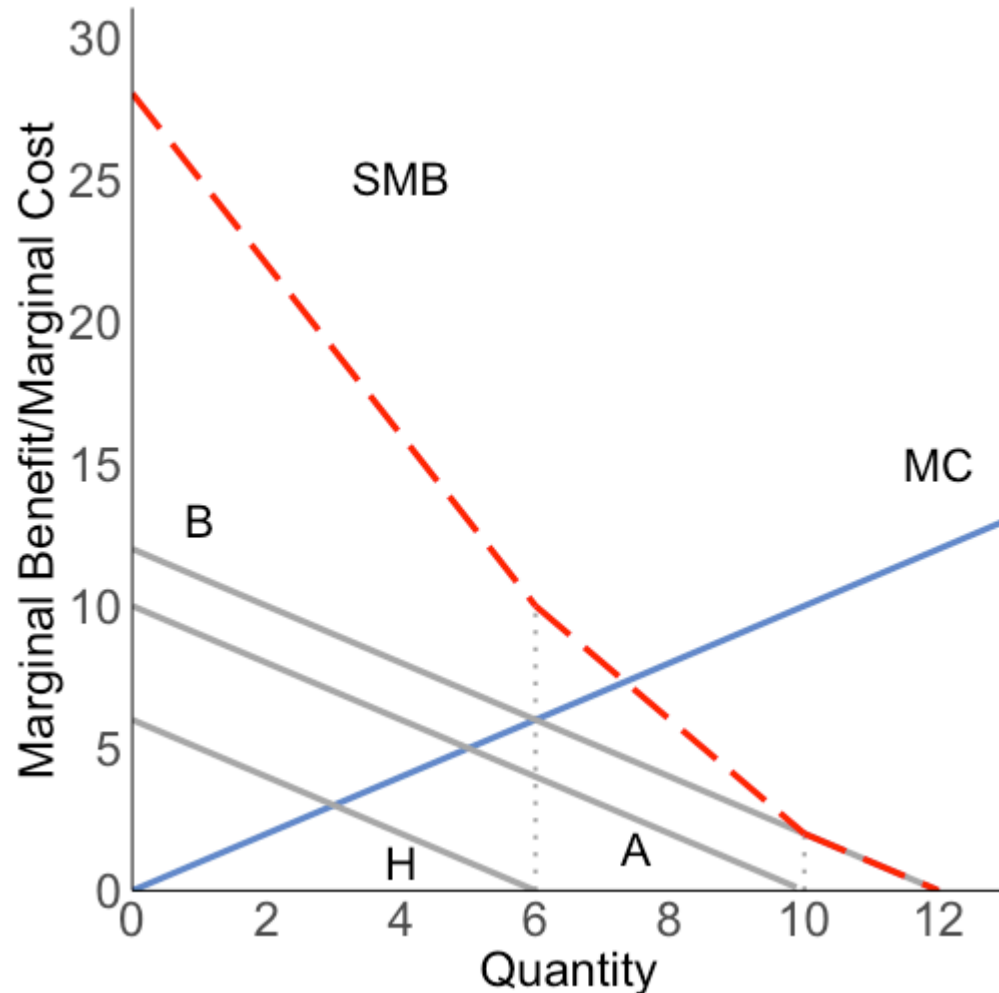


Why is the aggregate demand curve kinked?

Because at each quantity/depth, only certain groups are willing to pay

Kinks are at the dotted lines, where PMBs hit zero

Public goods: graphical

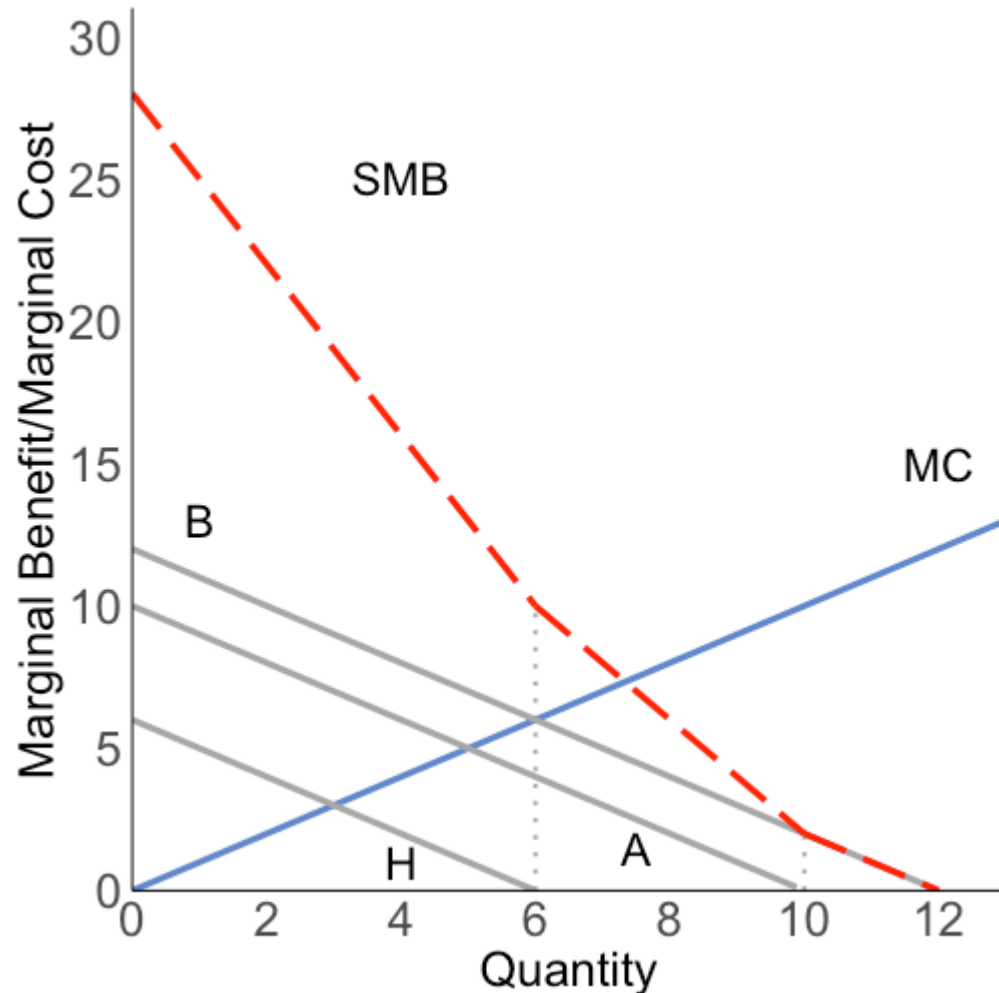


At quantities > 10 , only boaters are willing to pay

At quantities > 6 and ≤ 10 , only boaters and anglers are willing to pay

At quantities $Q \leq 6$, all groups are willing to pay

Public goods: graphical



The SMB curve is:

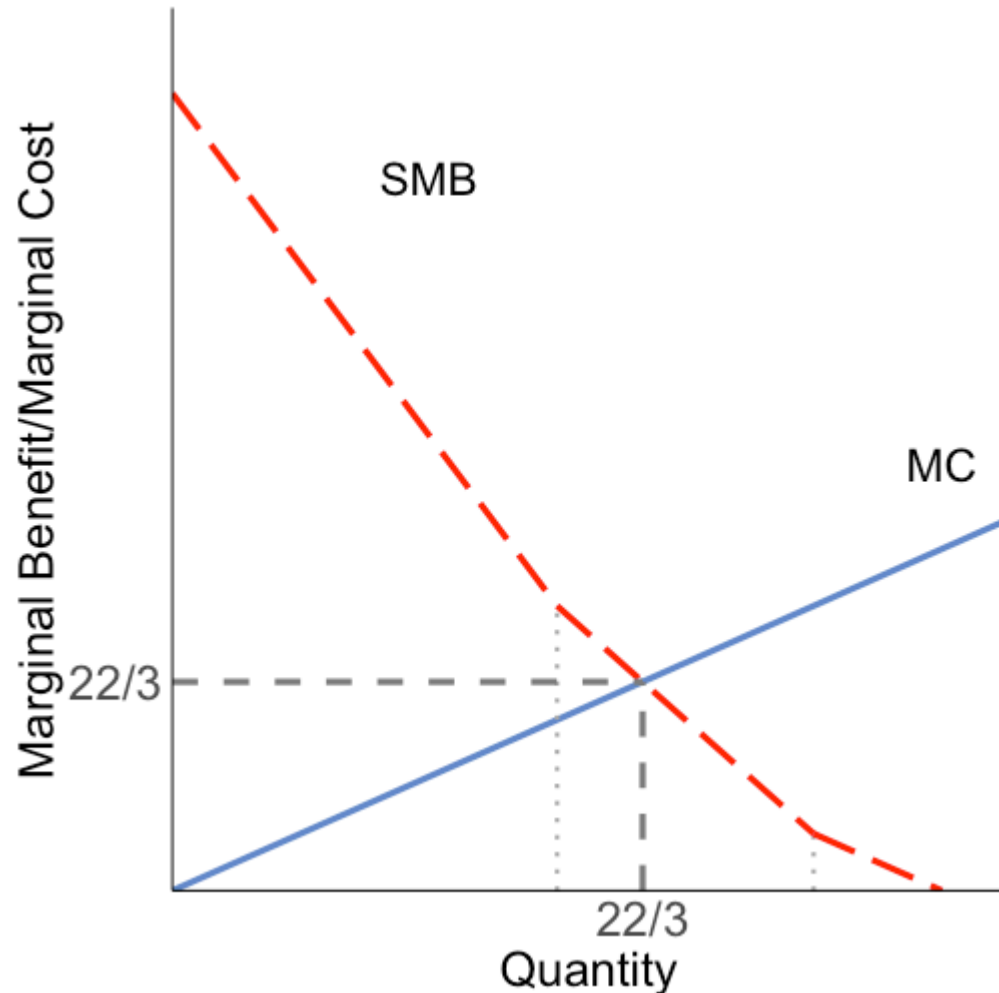
$28 - 3Q$ for $Q \leq 6$

$22 - 2Q$ for $6 < Q \leq 10$

$12 - Q$ for $10 < Q \leq 12$

Summing the PMBs over the relevant range of Q

Public goods: graphical



The optimal provision of the public good is where the MC curve crosses the SMB curve

This is across the middle segment

$$Q = 22 - 2Q \Rightarrow Q = 22/3$$

The optimal quantity of $Q = 22/3$ is greater than the quantity any individual group would be willing to purchase

Public goods financing

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If the government is able to provide the good, how does it finance the cost raising the river depth above zero?

Public goods financing

It can charge each group a constant share of this price

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Boaters pay: $14/3$

Anglers pay: $8/3$

Hikers: free

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Since the good is non-rival, this is enough to finance the cost

Public goods financing

What is a key problem with Lindahl pricing?

Public goods financing

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People can lie about which group they're in

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Anglers might say they're hikers

Public goods financing

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It requires perfect information on behalf of the regulator